Math 136 — Linear Algebra 1 for Honours Mathematics

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Contents

1 Vectors in $\mathbb{R}^n$ 1

1.1 Vectors and their operations . . . 1
1.2 Linear combinations and span . . . 2
1.3 Linear independence . . . 2
1.4 Bases of $\mathbb{R}^n$ and subspaces . . . 3
1.5 Lines and planes . . . 4
1.6 Dot product, length and angles . . . 4
1.7 The cross product in $\mathbb{R}^3$ . . . 5

2 Projections and Minimum Distance 6

2.1 Projection onto a vector . . . 6
2.2 Projection onto a plane . . . 7
2.3 Minimum distance problems . . . 7
2.4 Projection onto a subspace (preview) . . . 8

3 Systems of Linear Equations 9

3.1 Systems and solutions . . . 9
3.2 Augmented matrices and row operations . . . 9
3.3 Row echelon form and reduced row echelon form . . . 10
3.4 Gaussian and Gauss–Jordan elimination . . . 10
3.5 Rank and the structure of solution sets . . . 11
3.6 Homogeneous systems . . . 12

4 Matrices 13

4.1 Matrix addition and scalar multiplication . . . 13
4.2 Matrix–vector multiplication . . . 13
4.3 Matrix multiplication . . . 14
4.4 The identity matrix and special matrices . . . 15
4.5 Block multiplication . . . 15

5 Linear Mappings 16

5.1 Matrix mappings . . . 16
5.2 The standard matrix . . . 16
5.3 Geometric transformations . . . 17
5.4 Operations on linear mappings and composition . . . 17
5.5 Range and kernel . . . 18

6 The Four Fundamental Subspaces **20**
6.1 Definitions . . . 20
6.2 Dimension of a subspace of $\mathbb{R}^n$ . . . 20
6.3 Finding bases . . . 20
6.4 The Rank–Nullity Theorem . . . 22
6.5 Orthogonality of the fundamental subspaces . . . 22

7 Inverse Matrices and Elementary Matrices **23**
7.1 Inverses . . . 23
7.2 Computing inverses by row reduction . . . 24
7.3 Elementary matrices . . . 24
7.4 The Invertible Matrix Theorem . . . 25

8 Determinants **27**
8.1 Definition by cofactor expansion . . . 27
8.2 Determinants and row operations . . . 28
8.3 Products and transposes . . . 28
8.4 The adjugate and a formula for the inverse . . . 29
8.5 Cramer's Rule . . . 29
8.6 Area and volume . . . 30

9 Eigenvalues and Eigenvectors **31**
9.1 Definitions . . . 31
9.2 The characteristic polynomial . . . 31
9.3 Properties . . . 33

10 Diagonalization **35**
10.1 Similar matrices . . . 35
10.2 The Diagonalization Theorem . . . 35
10.3 Powers of matrices . . . 36

11 Vector Spaces **38**
11.1 The axioms . . . 38
11.2 Subspaces . . . 39
11.3 Span and linear independence . . . 39
11.4 Bases and dimension . . . 40
11.5 Coordinates . . . 40

12 General Linear Maps and Change of Basis **42**
12.1 Linear maps between vector spaces . . . 42
12.2 The matrix of a linear map . . . 43
12.3 Change of coordinates . . . 43
12.4 Change of basis for linear operators . . . 44
12.5 Isomorphisms . . . 45

Chapter 1

Vectors in $\mathbb{R}^n$

1.1 Vectors and their operations

Definition 1.1 ($\mathbb{R}^n$)**.** $\mathbb{R}^n$ is the set of all ordered n-tuples of real numbers, written as column vectors

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad x_1, \ldots, x_n \in \mathbb{R}.$$

The number x_i is the i-th *component* (or entry) of $\vec{x}$. Two vectors are equal when all their components agree. The *zero vector* is $\vec{0} = (0, \ldots, 0)^T$.

To save space we sometimes write $\vec{x} = (x_1, \ldots, x_n)$ or $\vec{x} = [x_1 \ \cdots \ x_n]^T$, but in this course vectors are *column* vectors.

Definition 1.2 (Addition and scalar multiplication)**.** For $\vec{x}, \vec{y} \in \mathbb{R}^n$ and $c \in \mathbb{R}$,

$$\vec{x} + \vec{y} = \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \qquad c\vec{x} = \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}.$$

We write $-\vec{x} = (-1)\vec{x}$ and $\vec{x} - \vec{y} = \vec{x} + (-\vec{y})$.

Theorem 1.3 (Properties of $\mathbb{R}^n$)**.** *For all $\vec{x}, \vec{y}, \vec{w} \in \mathbb{R}^n$ and $c, d \in \mathbb{R}$:*

V1. $\vec{x} + \vec{y} \in \mathbb{R}^n$ *(closed under addition);*

V2. $(\vec{x} + \vec{y}) + \vec{w} = \vec{x} + (\vec{y} + \vec{w})$*;*

V3. $\vec{x} + \vec{y} = \vec{y} + \vec{x}$*;*

V4. $\vec{x} + \vec{0} = \vec{x}$*;*

V5. $\vec{x} + (-\vec{x}) = \vec{0}$*;*

V6. $c\vec{x} \in \mathbb{R}^n$ *(closed under scalar multiplication);*

V7. $c(d\vec{x}) = (cd)\vec{x}$*;*

V8. $(c+d)\vec{x} = c\vec{x} + d\vec{x}$;

V9. $c(\vec{x}+\vec{y}) = c\vec{x} + c\vec{y}$;

V10. $1\vec{x} = \vec{x}$.

Proof. Each follows componentwise from the corresponding property of real numbers. For example for V8, the i-th component of $(c+d)\vec{x}$ is $(c+d)x_i = cx_i + dx_i$, which is the i-th component of $c\vec{x} + d\vec{x}$. □

These ten properties will later become the *axioms* of an abstract vector space (Chapter 11).

1.2 Linear combinations and span

Definition 1.4 (Linear combination)**.** For vectors $\vec{v}_1, \ldots, \vec{v}_k \in \mathbb{R}^n$ and scalars $c_1, \ldots, c_k \in \mathbb{R}$, the vector $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k$ is a *linear combination* of $\vec{v}_1, \ldots, \vec{v}_k$.

Definition 1.5 (Span)**.** The *span* of $B = \{\vec{v}_1, \ldots, \vec{v}_k\}$ is the set of all linear combinations:

$$\operatorname{Span} B = \{c_1\vec{v}_1 + \cdots + c_k\vec{v}_k : c_1, \ldots, c_k \in \mathbb{R}\}.$$

If $S = \operatorname{Span} B$ we say B *spans* S and that B is a *spanning set* for S. By convention $\operatorname{Span}\emptyset = \{\vec{0}\}$.

Example 1.6. Is $\vec{x} = (1,2,3)$ in $\operatorname{Span}\{(1,0,1),(0,1,1)\}$? We need $c_1(1,0,1) + c_2(0,1,1) = (c_1, c_2, c_1+c_2) = (1,2,3)$. The first two components force $c_1 = 1$, $c_2 = 2$, and then $c_1 + c_2 = 3$ agrees with the third component. So yes: $\vec{x} = 1(1,0,1) + 2(0,1,1)$. On the other hand $(1,2,4)$ is *not* in the span, since $c_1 + c_2 = 3 \neq 4$.

Theorem 1.7 (Removing redundant vectors)**.** *Let $\vec{v}_1, \ldots, \vec{v}_k \in \mathbb{R}^n$. If $\vec{v}_k$ is a linear combination of $\vec{v}_1, \ldots, \vec{v}_{k-1}$, then* $\operatorname{Span}\{\vec{v}_1, \ldots, \vec{v}_k\} = \operatorname{Span}\{\vec{v}_1, \ldots, \vec{v}_{k-1}\}$.

Proof. The inclusion $\supseteq$ is clear (take coefficient 0 on $\vec{v}_k$). Conversely, write $\vec{v}_k = a_1\vec{v}_1 + \cdots + a_{k-1}\vec{v}_{k-1}$. Then any $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = (c_1 + c_k a_1)\vec{v}_1 + \cdots + (c_{k-1} + c_k a_{k-1})\vec{v}_{k-1}$ lies in the smaller span. □

1.3 Linear independence

Definition 1.8 (Linear (in)dependence)**.** A set $\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \mathbb{R}^n$ is *linearly dependent* if there are scalars $c_1, \ldots, c_k$, *not all zero*, with $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$. Otherwise it is *linearly independent*: the only solution of $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$ is the *trivial solution* $c_1 = \cdots = c_k = 0$.

Theorem 1.9. *A set $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is linearly dependent if and only if some $\vec{v}_i$ is a linear combination of the others. In particular, any set containing $\vec{0}$ is linearly dependent.*

Proof. If $c_1\vec{v}_1 + \cdots + c_k\vec{v}_k = \vec{0}$ with $c_i \neq 0$, then $\vec{v}_i = -\frac{1}{c_i}\sum_{j\neq i} c_j\vec{v}_j$. Conversely, if $\vec{v}_i = \sum_{j\neq i} a_j\vec{v}_j$, then $\sum_{j\neq i} a_j\vec{v}_j - 1\vec{v}_i = \vec{0}$ is a nontrivial relation (coefficient of $\vec{v}_i$ is $-1 \neq 0$). If $\vec{v}_1 = \vec{0}$, then $1\vec{v}_1 + 0\vec{v}_2 + \cdots = \vec{0}$. □

Example 1.10. Is $\{(1,2,1),(0,1,1),(1,3,2)\}$ linearly independent? Consider $c_1(1,2,1)+c_2(0,1,1)+c_3(1,3,2)=\vec{0}$, i.e.

$$c_1+c_3=0, \qquad 2c_1+c_2+3c_3=0, \qquad c_1+c_2+2c_3=0.$$

From the first, $c_1=-c_3$; substituting into the third gives $c_2=-c_3$; the second becomes $-2c_3-c_3+3c_3=0$, which always holds. Taking $c_3=1$ gives the nontrivial relation $-(1,2,1)-(0,1,1)+(1,3,2)=\vec{0}$, so the set is linearly *dependent*. Indeed $(1,3,2)=(1,2,1)+(0,1,1)$.

In Chapter 3 we will solve such systems systematically by row reduction.

1.4 Bases of $\mathbb{R}^n$ and subspaces

Definition 1.11 (Basis)**.** A set B of vectors in a subspace S (or in $\mathbb{R}^n$) is a *basis* for S if B is linearly independent and $\operatorname{Span} B = S$.

Example 1.12 (Standard basis)**.** The *standard basis* of $\mathbb{R}^n$ is $\{\vec{e}_1,\dots,\vec{e}_n\}$, where $\vec{e}_i$ has a 1 in position i and 0 elsewhere. Every $\vec{x}$ is $x_1\vec{e}_1+\cdots+x_n\vec{e}_n$, and $c_1\vec{e}_1+\cdots+c_n\vec{e}_n=(c_1,\dots,c_n)=\vec{0}$ forces all $c_i=0$.

Theorem 1.13 (Unique representation)**.** *If $B=\{\vec{v}_1,\dots,\vec{v}_k\}$ is a basis for S, then every $\vec{x}\in S$ can be written as a linear combination of the vectors of B in* exactly one *way.*

Proof. Existence holds because B spans S. If $\vec{x}=\sum c_i\vec{v}_i=\sum d_i\vec{v}_i$, subtracting gives $\sum(c_i-d_i)\vec{v}_i=\vec{0}$, so $c_i=d_i$ for all i by independence. □

We will see (Chapter 3) that a set of k vectors in $\mathbb{R}^n$ can only be linearly independent if $k\le n$ and can only span $\mathbb{R}^n$ if $k\ge n$; hence every basis of $\mathbb{R}^n$ has exactly n vectors, and *a set of n vectors in $\mathbb{R}^n$ is a basis iff it is linearly independent iff it spans $\mathbb{R}^n$.*

Definition 1.14 (Subspace)**.** A subset $S\subseteq\mathbb{R}^n$ is a *subspace* of $\mathbb{R}^n$ if S together with the operations of $\mathbb{R}^n$ satisfies all of V1–V10.

Most of the properties are inherited automatically from $\mathbb{R}^n$, which gives a short test.

Theorem 1.15 (Subspace Test)**.** *A subset $S\subseteq\mathbb{R}^n$ is a subspace if and only if*

1. *S is non-empty (equivalently, $\vec{0}\in S$),*
2. *$\vec{x}+\vec{y}\in S$ for all $\vec{x},\vec{y}\in S$, and*
3. *$c\vec{x}\in S$ for all $\vec{x}\in S$, $c\in\mathbb{R}$.*

Proof sketch. V2, V3, V7–V10 hold for vectors in S because they hold in $\mathbb{R}^n$. V1 and V6 are conditions 2 and 3. For V4 and V5: pick $\vec{x}\in S$; then $0\vec{x}=\vec{0}\in S$ and $(-1)\vec{x}=-\vec{x}\in S$ by condition 3. □

Example 1.16. $S=\{\vec{x}\in\mathbb{R}^3 : x_1+x_2-x_3=0\}$ is a subspace. (1) $0+0-0=0$, so $\vec{0}\in S$. (2) If $\vec{x},\vec{y}\in S$ then $(x_1+y_1)+(x_2+y_2)-(x_3+y_3)=(x_1+x_2-x_3)+(y_1+y_2-y_3)=0$. (3) $cx_1+cx_2-cx_3=c(x_1+x_2-x_3)=0$. Solving $x_3=x_1+x_2$ shows $S=\operatorname{Span}\{(1,0,1),(0,1,1)\}$, and these two vectors are clearly independent, so they form a basis of S.

Example 1.17. $T=\{\vec{x}\in\mathbb{R}^2 : x_1=1\}$ is *not* a subspace since $\vec{0}\notin T$. And $U=\{\vec{x}\in\mathbb{R}^2 : x_1x_2=0\}$ (the two axes) is not a subspace: $(1,0),(0,1)\in U$ but $(1,1)\notin U$.

Theorem 1.18. *For any $\vec{v}_1,\dots,\vec{v}_k\in\mathbb{R}^n$, $\operatorname{Span}\{\vec{v}_1,\dots,\vec{v}_k\}$ is a subspace of $\mathbb{R}^n$.*

Proof. $\vec{0}=0\vec{v}_1+\cdots+0\vec{v}_k$ is in the span; sums and scalar multiples of linear combinations are again linear combinations. □

1.5 Lines and planes

Definition 1.19. Let $\vec{p}, \vec{d} \in \mathbb{R}^n$ with $\vec{d} \neq \vec{0}$. The set with *vector equation* $\vec{x} = \vec{p} + t\vec{d}$, $t \in \mathbb{R}$, is the *line* through $\vec{p}$ with direction $\vec{d}$. If $\{\vec{u}, \vec{v}\}$ is linearly independent, $\vec{x} = \vec{p} + s\vec{u} + t\vec{v}$, $s, t \in \mathbb{R}$, is the *plane* through $\vec{p}$ spanned by $\vec{u}, \vec{v}$.

Geometrically, the span of one non-zero vector is a line through the origin, and the span of two independent vectors is a plane through the origin. A line or plane that does not pass through the origin is a *translate* of a subspace, not a subspace itself.

Example 1.20. The line through $P(1, 2, -1)$ and $Q(3, 0, 4)$ has direction $\vec{PQ} = (2, -2, 5)$, so a vector equation is $\vec{x} = (1, 2, -1) + t(2, -2, 5)$, $t \in \mathbb{R}$. Its *parametric equations* are $x_1 = 1 + 2t$, $x_2 = 2 - 2t$, $x_3 = -1 + 5t$.

1.6 Dot product, length and angles

Definition 1.21 (Dot product, length)**.** For $\vec{x}, \vec{y} \in \mathbb{R}^n$, the *dot product* is $\vec{x} \cdot \vec{y} = x_1y_1 + \cdots + x_ny_n$, and the *length* (norm) of $\vec{x}$ is $\|\vec{x}\| = \sqrt{\vec{x} \cdot \vec{x}}$. A vector of length 1 is a *unit vector*; for $\vec{x} \neq \vec{0}$, $\hat{x} = \frac{1}{\|\vec{x}\|}\vec{x}$ is the unit vector in the direction of $\vec{x}$.

Theorem 1.22 (Properties of the dot product)**.** *For $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$, $c \in \mathbb{R}$: (1) $\vec{x} \cdot \vec{x} \geq 0$ with equality iff $\vec{x} = \vec{0}$; (2) $\vec{x} \cdot \vec{y} = \vec{y} \cdot \vec{x}$; (3) $\vec{x} \cdot (\vec{y} + \vec{z}) = \vec{x} \cdot \vec{y} + \vec{x} \cdot \vec{z}$; (4) $(c\vec{x}) \cdot \vec{y} = c(\vec{x} \cdot \vec{y})$; (5) $\|c\vec{x}\| = |c| \, \|\vec{x}\|$; (6) $|\vec{x} \cdot \vec{y}| \leq \|\vec{x}\| \, \|\vec{y}\|$ (Cauchy–Schwarz); (7) $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$ (triangle inequality).*

Proof of (6) and (7). If $\vec{y} = \vec{0}$ both sides of (6) are 0. Otherwise, for every $t \in \mathbb{R}$,

$$0 \leq \|\vec{x} + t\vec{y}\|^2 = \|\vec{y}\|^2 t^2 + 2(\vec{x} \cdot \vec{y})t + \|\vec{x}\|^2 .$$

A quadratic in t that is never negative has discriminant ≤ 0: $4(\vec{x} \cdot \vec{y})^2 - 4\|\vec{x}\|^2 \|\vec{y}\|^2 \leq 0$, which is (6). For (7), $\|\vec{x} + \vec{y}\|^2 = \|\vec{x}\|^2 + 2\vec{x} \cdot \vec{y} + \|\vec{y}\|^2 \leq \|\vec{x}\|^2 + 2\|\vec{x}\| \, \|\vec{y}\| + \|\vec{y}\|^2 = (\|\vec{x}\| + \|\vec{y}\|)^2$. □

By Cauchy–Schwarz, $-1 \leq \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} \leq 1$ for non-zero vectors, so we can *define* the angle $\theta \in [0, \pi]$ between them by

$$\cos\theta = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \, \|\vec{y}\|}.$$

In $\mathbb{R}^2$ and $\mathbb{R}^3$ this agrees with the usual angle (law of cosines).

Example 1.23. Let $\vec{u} = (1, 2, 2)$, $\vec{v} = (3, 0, 4)$. Then $\vec{u} \cdot \vec{v} = 3 + 0 + 8 = 11$, $\|\vec{u}\| = \sqrt{1 + 4 + 4} = 3$, $\|\vec{v}\| = \sqrt{9 + 0 + 16} = 5$, so $\cos\theta = 11/15$ and $\theta = \arccos(11/15) \approx 0.75$ rad.

Definition 1.24 (Orthogonal)**.** $\vec{x}, \vec{y} \in \mathbb{R}^n$ are *orthogonal* if $\vec{x} \cdot \vec{y} = 0$. (So $\vec{0}$ is orthogonal to everything.) A set of vectors is *orthogonal* if every pair of distinct vectors in it is orthogonal.

Definition 1.25 (Scalar equation of a plane)**.** In $\mathbb{R}^3$, a non-zero vector $\vec{n}$ is a *normal vector* of a plane if it is orthogonal to every direction vector in the plane. If the plane passes through P, then X is on the plane iff $\vec{n} \cdot (\vec{x} - \vec{p}) = 0$. With $\vec{n} = (a, b, c)$ this is the scalar equation $ax_1 + bx_2 + cx_3 = \vec{n} \cdot \vec{p}$. More generally $\vec{n} \cdot \vec{x} = d$ in $\mathbb{R}^n$ defines a *hyperplane*.

1.7 The cross product in $\mathbb{R}^3$

Definition 1.26 (Cross product)**.** For $\vec{u} = (u_1, u_2, u_3)$, $\vec{v} = (v_1, v_2, v_3)$ in $\mathbb{R}^3$,

$$\vec{u} \times \vec{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}.$$

Theorem 1.27. *For $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$ and $c \in \mathbb{R}$:*

1. *$\vec{u} \times \vec{v}$ is orthogonal to both $\vec{u}$ and $\vec{v}$;*
2. *$\vec{v} \times \vec{u} = -(\vec{u} \times \vec{v})$, so $\vec{u} \times \vec{u} = \vec{0}$;*
3. *$\vec{u} \times \vec{v} = \vec{0}$ iff one of $\vec{u}, \vec{v}$ is a scalar multiple of the other;*
4. *$\vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$ and $(c\vec{u}) \times \vec{v} = c(\vec{u} \times \vec{v})$;*
5. *$\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \, \|\vec{v}\| \sin\theta$, which is the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$.*

Proof of (1). $\vec{u} \cdot (\vec{u} \times \vec{v}) = u_1(u_2 v_3 - u_3 v_2) + u_2(u_3 v_1 - u_1 v_3) + u_3(u_1 v_2 - u_2 v_1)$; the six terms cancel in pairs. Similarly for $\vec{v}$. □

Pitfall

The cross product is *not* associative: $\vec{e}_1 \times (\vec{e}_1 \times \vec{e}_2) = \vec{e}_1 \times \vec{e}_3 = -\vec{e}_2$, but $(\vec{e}_1 \times \vec{e}_1) \times \vec{e}_2 = \vec{0}$. It is only defined in $\mathbb{R}^3$.

Example 1.28. $(1,2,3) \times (4,5,6) = (2 \cdot 6 - 3 \cdot 5,\ 3 \cdot 4 - 1 \cdot 6,\ 1 \cdot 5 - 2 \cdot 4) = (-3, 6, -3)$. Check: $(1,2,3) \cdot (-3,6,-3) = -3 + 12 - 9 = 0$ and $(4,5,6) \cdot (-3,6,-3) = -12 + 30 - 18 = 0$.

Example 1.29 (Plane through three points)**.** Find the scalar equation of the plane through $P(1,0,0)$, $Q(0,2,0)$, $R(0,0,3)$. Direction vectors are $\vec{PQ} = (-1,2,0)$ and $\vec{PR} = (-1,0,3)$, so a normal is

$$\vec{n} = \vec{PQ} \times \vec{PR} = (2 \cdot 3 - 0 \cdot 0,\ 0 \cdot (-1) - (-1) \cdot 3,\ (-1) \cdot 0 - 2 \cdot (-1)) = (6, 3, 2).$$

The plane is $6x_1 + 3x_2 + 2x_3 = \vec{n} \cdot \vec{p} = 6$. Check: Q gives 6, R gives 6. The triangle PQR has area $\frac{1}{2}\|\vec{n}\| = \frac{1}{2}\sqrt{36 + 9 + 4} = \frac{7}{2}$.

Summary

- $\mathbb{R}^n$ with componentwise $+$ and scalar multiplication satisfies ten properties V1–V10.
- Span = all linear combinations; it is always a subspace. A redundant vector can be removed without changing the span.
- Linearly independent = only the trivial relation; dependent iff some vector is a combination of the others. A basis is an independent spanning set and gives unique coordinates.
- Subspace test: contains $\vec{0}$, closed under $+$ and scalar multiplication.
- $\vec{x} \cdot \vec{y}$, $\|\vec{x}\|$, $\cos\theta = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\|\|\vec{y}\|}$; Cauchy–Schwarz and triangle inequality; orthogonal means dot product 0.
- $\vec{u} \times \vec{v}$ is orthogonal to both, has length equal to the parallelogram's area, and gives normal vectors of planes in $\mathbb{R}^3$.

Chapter 2

Projections and Minimum Distance

2.1 Projection onto a vector

A basic geometric problem: given $\vec{u}$ and a non-zero $\vec{v}$ in $\mathbb{R}^n$, split $\vec{u}$ into a piece parallel to $\vec{v}$ and a piece orthogonal to $\vec{v}$. We look for $\vec{u} = c\vec{v} + \vec{w}$ with $\vec{w} \cdot \vec{v} = 0$. Dotting both sides with $\vec{v}$ gives $\vec{u} \cdot \vec{v} = c\,\|\vec{v}\|^2$, so c is forced to be $\frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|^2}$.

Definition 2.1 (Projection and perpendicular)**.** For $\vec{u}, \vec{v} \in \mathbb{R}^n$ with $\vec{v} \neq \vec{0}$, the *projection of $\vec{u}$ onto $\vec{v}$* is

$$\operatorname{proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2}\, \vec{v},$$

and the *perpendicular of $\vec{u}$ onto $\vec{v}$* is $\operatorname{perp}_{\vec{v}} \vec{u} = \vec{u} - \operatorname{proj}_{\vec{v}} \vec{u}$.

Theorem 2.2 (Orthogonal decomposition)**.** *For $\vec{v} \neq \vec{0}$: $\vec{u} = \operatorname{proj}_{\vec{v}} \vec{u} + \operatorname{perp}_{\vec{v}} \vec{u}$, where $\operatorname{proj}_{\vec{v}} \vec{u}$ is a scalar multiple of $\vec{v}$ and $\operatorname{perp}_{\vec{v}} \vec{u}$ is orthogonal to $\vec{v}$. This decomposition is unique.*

Proof. $\operatorname{perp}_{\vec{v}} \vec{u} \cdot \vec{v} = \vec{u} \cdot \vec{v} - \frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|^2} \vec{v} \cdot \vec{v} = \vec{u} \cdot \vec{v} - \vec{u} \cdot \vec{v} = 0$. Uniqueness was shown in the derivation above: c is forced. □

Example 2.3. Let $\vec{u} = (2, 3, 1)$ and $\vec{v} = (1, 1, 0)$. Then $\vec{u} \cdot \vec{v} = 2 + 3 + 0 = 5$ and $\|\vec{v}\|^2 = 2$, so

$$\operatorname{proj}_{\vec{v}} \vec{u} = \tfrac{5}{2}(1, 1, 0) = \left(\tfrac{5}{2}, \tfrac{5}{2}, 0\right), \qquad \operatorname{perp}_{\vec{v}} \vec{u} = (2, 3, 1) - \left(\tfrac{5}{2}, \tfrac{5}{2}, 0\right) = \left(-\tfrac{1}{2}, \tfrac{1}{2}, 1\right).$$

Check: $\left(-\frac{1}{2}, \frac{1}{2}, 1\right) \cdot (1, 1, 0) = -\frac{1}{2} + \frac{1}{2} + 0 = 0$.

Theorem 2.4 (Properties of proj)**.** *For $\vec{v} \neq \vec{0}$, $\vec{x}, \vec{y} \in \mathbb{R}^n$, $c \in \mathbb{R}$:*

1. $\operatorname{proj}_{\vec{v}}(\vec{x} + \vec{y}) = \operatorname{proj}_{\vec{v}} \vec{x} + \operatorname{proj}_{\vec{v}} \vec{y}$ *and* $\operatorname{proj}_{\vec{v}}(c\vec{x}) = c \operatorname{proj}_{\vec{v}} \vec{x}$ *(it is linear; see Chapter 5);*
2. $\operatorname{proj}_{\vec{v}}(\operatorname{proj}_{\vec{v}} \vec{x}) = \operatorname{proj}_{\vec{v}} \vec{x}$ *(projecting twice changes nothing);*
3. $\operatorname{proj}_{\vec{v}}(\operatorname{perp}_{\vec{v}} \vec{x}) = \vec{0}$*;*
4. $\operatorname{proj}_{c\vec{v}} \vec{x} = \operatorname{proj}_{\vec{v}} \vec{x}$ *for $c \neq 0$ (only the line matters).*

Pitfall

$\operatorname{proj}_{\vec{v}} \vec{u}$ and $\operatorname{proj}_{\vec{u}} \vec{v}$ are generally different: the first is a multiple of $\vec{v}$, the second a multiple of $\vec{u}$. Also, the denominator is $\|\vec{v}\|^2$, not $\|\vec{v}\|$. If $\vec{v}$ is a unit vector, the formula simplifies to $(\vec{u} \cdot \vec{v})\vec{v}$.

2.2 Projection onto a plane

For a plane through the origin in $\mathbb{R}^3$ with normal $\vec{n}$, the part of $\vec{x}$ *in* the plane is what remains after removing the component along $\vec{n}$:

$$\operatorname{proj}_P \vec{x} = \operatorname{perp}_{\vec{n}} \vec{x} = \vec{x} - \frac{\vec{x}\cdot\vec{n}}{\|\vec{n}\|^2}\vec{n}.$$

Example 2.5. Project $\vec{x} = (1,1,1)$ onto the plane $P : x_1 + 2x_2 - x_3 = 0$. Here $\vec{n} = (1,2,-1)$, $\vec{x}\cdot\vec{n} = 1+2-1 = 2$, $\|\vec{n}\|^2 = 6$, so

$$\operatorname{proj}_{\vec{n}} \vec{x} = \tfrac{2}{6}(1,2,-1) = \left(\tfrac{1}{3},\tfrac{2}{3},-\tfrac{1}{3}\right), \qquad \operatorname{proj}_P \vec{x} = (1,1,1) - \left(\tfrac{1}{3},\tfrac{2}{3},-\tfrac{1}{3}\right) = \left(\tfrac{2}{3},\tfrac{1}{3},\tfrac{4}{3}\right).$$

Check it lies in P: $\frac{2}{3} + 2\cdot\frac{1}{3} - \frac{4}{3} = 0$.

2.3 Minimum distance problems

Theorem 2.6. *Let $\vec{v} \neq \vec{0}$. Among all points $t\vec{v}$ on the line* $\operatorname{Span}\{\vec{v}\}$*, the one closest to $\vec{u}$ is* $\operatorname{proj}_{\vec{v}} \vec{u}$*, and the minimum distance is* $\|\operatorname{perp}_{\vec{v}} \vec{u}\|$.

Proof. Write $\vec{w} = \operatorname{perp}_{\vec{v}} \vec{u}$, so $\vec{u} - t\vec{v} = \vec{w} + (\operatorname{proj}_{\vec{v}} \vec{u} - t\vec{v})$, where the second term is a multiple of $\vec{v}$, hence orthogonal to $\vec{w}$. By Pythagoras, $\|\vec{u} - t\vec{v}\|^2 = \|\vec{w}\|^2 + \|\operatorname{proj}_{\vec{v}} \vec{u} - t\vec{v}\|^2 \geq \|\vec{w}\|^2$, with equality exactly when $t\vec{v} = \operatorname{proj}_{\vec{v}} \vec{u}$. □

The same idea handles lines and planes not through the origin: translate so that a known point of the object becomes the origin.

Recipes

Point Q to line $\vec{x} = \vec{p} + t\vec{d}$: let $\vec{PQ} = \vec{q} - \vec{p}$. The distance is $\left\|\operatorname{perp}_{\vec{d}} \vec{PQ}\right\|$; the closest point is $\vec{p} + \operatorname{proj}_{\vec{d}} \vec{PQ}$.
Point Q to plane $\vec{n}\cdot\vec{x} = d$ in $\mathbb{R}^3$: pick any point P on the plane. The distance is $\left\|\operatorname{proj}_{\vec{n}} \vec{PQ}\right\| = \dfrac{|\vec{n}\cdot\vec{q} - d|}{\|\vec{n}\|}$; the closest point is $\vec{q} - \operatorname{proj}_{\vec{n}} \vec{PQ}$.

Example 2.7 (Point to line)**.** Find the distance from $Q(3,1,2)$ to the line $\vec{x} = (1,0,0) + t(1,1,1)$. Here $\vec{PQ} = (2,1,2)$, $\vec{d} = (1,1,1)$, $\vec{PQ}\cdot\vec{d} = 5$, $\left\|\vec{d}\right\|^2 = 3$, so

$$\operatorname{proj}_{\vec{d}} \vec{PQ} = \tfrac{5}{3}(1,1,1), \qquad \operatorname{perp}_{\vec{d}} \vec{PQ} = (2,1,2) - \left(\tfrac{5}{3},\tfrac{5}{3},\tfrac{5}{3}\right) = \left(\tfrac{1}{3},-\tfrac{2}{3},\tfrac{1}{3}\right).$$

The distance is $\sqrt{\frac{1}{9}+\frac{4}{9}+\frac{1}{9}} = \frac{\sqrt{6}}{3}$, and the closest point is $(1,0,0) + \frac{5}{3}(1,1,1) = \left(\frac{8}{3},\frac{5}{3},\frac{5}{3}\right)$. Check: $Q - \left(\frac{8}{3},\frac{5}{3},\frac{5}{3}\right) = \left(\frac{1}{3},-\frac{2}{3},\frac{1}{3}\right)$, which is orthogonal to $\vec{d}$.

Example 2.8 (Point to plane)**.** Find the point on the plane $2x_1 - x_2 + 2x_3 = 5$ closest to $Q(1,2,3)$. Take $P(0,-5,0)$ on the plane (check: $0 + 5 + 0 = 5$). Then $\vec{PQ} = (1,7,3)$, $\vec{n} = (2,-1,2)$, $\vec{PQ}\cdot\vec{n} = 2 - 7 + 6 = 1$, $\|\vec{n}\|^2 = 9$. So

$$\operatorname{proj}_{\vec{n}} \vec{PQ} = \tfrac{1}{9}(2,-1,2), \qquad \text{distance} = \tfrac{1}{9}\|\vec{n}\| = \tfrac{1}{3},$$

and the closest point is $R = (1,2,3) - \frac{1}{9}(2,-1,2) = \left(\frac{7}{9},\frac{19}{9},\frac{25}{9}\right)$. Check: $2\cdot\frac{7}{9} - \frac{19}{9} + 2\cdot\frac{25}{9} = \frac{14-19+50}{9} = \frac{45}{9} = 5$. Also the shortcut formula gives $|2 - 2 + 6 - 5|/3 = 1/3$.

Example 2.9 (Distance between parallel planes)**.** The planes $x_1 + 2x_2 + 2x_3 = 1$ and $x_1 + 2x_2 + 2x_3 = 7$ are parallel (same normal $(1, 2, 2)$). The point $(1, 0, 0)$ is on the first, so the distance is $|1 - 7| \,/\, \|(1, 2, 2)\| = 6/3 = 2$.

Skew lines. For two non-parallel lines $\vec{x} = \vec{p} + s\vec{d_1}$, $\vec{x} = \vec{q} + t\vec{d_2}$ in $\mathbb{R}^3$, the vector $\vec{n} = \vec{d_1} \times \vec{d_2}$ is orthogonal to both, and the distance between the lines is $\|\text{proj}_{\vec{n}}(\vec{q} - \vec{p})\|$.

2.4 Projection onto a subspace (preview)

If $\{\vec{v}_1, \ldots, \vec{v}_k\}$ is an *orthogonal* basis of a subspace S, then the point of S closest to $\vec{x}$ is

$$\text{proj}_S \vec{x} = \text{proj}_{\vec{v}_1} \vec{x} + \cdots + \text{proj}_{\vec{v}_k} \vec{x},$$

and $\vec{x} - \text{proj}_S \vec{x}$ is orthogonal to every vector in S. Constructing orthogonal bases (Gram–Schmidt) and the general theory of orthogonal projections is done in Math 235; here we only need the one-vector and plane cases.

Summary

- $\text{proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v}$ and $\text{perp}_{\vec{v}} \vec{u} = \vec{u} - \text{proj}_{\vec{v}} \vec{u}$ give the unique decomposition of $\vec{u}$ into parts parallel and orthogonal to $\vec{v}$.
- Projection onto a plane through $\vec{0}$ with normal $\vec{n}$ is $\text{perp}_{\vec{n}}$.
- The closest point on a line/plane is found by a projection; distances are lengths of the perpendicular part. Point-to-plane distance is $|\vec{n} \cdot \vec{q} - d| \,/\, \|\vec{n}\|$.

Chapter 3

Systems of Linear Equations

3.1 Systems and solutions

Definition 3.1. A *linear equation* in variables $x_1, \ldots, x_n$ is one of the form $a_1x_1 + \cdots + a_nx_n = b$ with $a_i, b \in \mathbb{R}$. A *system of linear equations* is a finite collection of such equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

A *solution* is a vector $\vec{s} \in \mathbb{R}^n$ satisfying all m equations simultaneously. The *solution set* is the set of all solutions. The system is *consistent* if it has at least one solution and *inconsistent* otherwise.

Theorem 3.2. *A system of linear equations has either no solutions, exactly one solution, or infinitely many solutions.*

Proof. Suppose $\vec{s} \neq \vec{t}$ are both solutions. For each $c \in \mathbb{R}$ consider $\vec{s} + c(\vec{s} - \vec{t})$. Plugging into equation i: $\sum_j a_{ij}(s_j + c(s_j - t_j)) = b_i + c(b_i - b_i) = b_i$. Distinct values of c give distinct vectors, so there are infinitely many solutions. □

3.2 Augmented matrices and row operations

We record only the numbers. The *coefficient matrix* is $A = [a_{ij}]$ $(m \times n)$ and the *augmented matrix* is $[A \mid \vec{b}]$:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ \vdots & & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right].$$

Definition 3.3 (Elementary row operations)**.**

1. Multiply a row by a *non-zero* scalar: cR_i, $c \neq 0$.
2. Add a multiple of one row to another: $R_i + cR_j$ $(i \neq j)$.
3. Swap two rows: $R_i \leftrightarrow R_j$.

Two matrices are *row equivalent* if one can be obtained from the other by a finite sequence of elementary row operations.

Theorem 3.4. *If* $[A \mid \vec{b}]$ *and* $[C \mid \vec{d}]$ *are row equivalent, the corresponding systems have the same solution set.*

Proof sketch. Any solution of the original equations satisfies the new equations (they are combinations of the old ones). Each row operation is reversible by an operation of the same type ($\frac{1}{c}R_i$, $R_i - cR_j$, $R_i \leftrightarrow R_j$), so the converse holds too. □

3.3 Row echelon form and reduced row echelon form

Definition 3.5 (REF and RREF)**.** A matrix is in *row echelon form* (REF) if

1. all zero rows are at the bottom, and
2. the first non-zero entry (the *leading entry*) of each non-zero row is strictly to the right of the leading entry of the row above.

It is in *reduced row echelon form* (RREF) if additionally

3. every leading entry is 1 (a *leading one*), and
4. each leading one is the only non-zero entry in its column.

Theorem 3.6. *Every matrix is row equivalent to a matrix in REF, and to a* unique *matrix in RREF.*

Uniqueness of the RREF is what makes the next definition meaningful (an REF is not unique, but the positions of its leading entries are).

Definition 3.7 (Rank, pivots)**.** The *rank* of A, $\operatorname{rank} A$, is the number of leading ones in the RREF of A (equivalently, the number of leading entries in any REF). Columns containing leading entries are *pivot columns*; in a system, the variables of pivot columns are *leading variables* and the rest are *free variables*.

3.4 Gaussian and Gauss–Jordan elimination

Algorithm

Gaussian elimination: row reduce $[A \mid \vec{b}]$ to REF working left to right, creating zeros *below* each leading entry; then solve by back-substitution. **Gauss–Jordan elimination**: continue to RREF (zeros above and below every leading one), then read the solution off directly.

Example 3.8 (A unique solution)**.** Solve $x_1 + x_2 + x_3 = 6$, $2x_1 + 3x_2 + x_3 = 11$, $x_1 - x_2 + 2x_3 = 5$.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 1 & 11 \\ 1 & -1 & 2 & 5 \end{array}\right] \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & -2 & 1 & -1 \end{array}\right] \xrightarrow{R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -1 & -3 \end{array}\right]$$

This is an REF. Continuing to RREF:

$$\xrightarrow{-R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right] \xrightarrow[R_2 + R_3]{R_1 - R_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right] \xrightarrow{R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right].$$

The unique solution is $\vec{x} = (1, 2, 3)$. Check in the second equation: $2 + 6 + 3 = 11$.

Example 3.9 (Infinitely many solutions; parametric vector form)**.** Solve the system with augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 1 & 4 & 8 \\ 1 & 2 & 1 & 3 & 5 \end{array}\right] \xrightarrow[R_3-R_1]{R_2-2R_1} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 2 \end{array}\right] \xrightarrow{R_3-R_2} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

This is already RREF. Leading variables: x_1, x_3; free variables: $x_2 = s$, $x_4 = t$. Then $x_1 = 3-2s-t$, $x_3 = 2-2t$, and the solution set in *parametric vector form* is

$$\vec{x} = \begin{bmatrix} 3 \\ 0 \\ 2 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \qquad s, t \in \mathbb{R}.$$

Check with the third original equation: $(3-2s-t)+2s+(2-2t)+3t = 5$. Geometrically this is a plane in $\mathbb{R}^4$ not through the origin.

Example 3.10 (Depending on parameters)**.** For which k, m is the system with augmented matrix below inconsistent / uniquely solvable / infinitely solvable?

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 2 \\ 1 & 3 & k & m \end{array}\right] \xrightarrow[R_3-R_1]{R_2-R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 2 & k-1 & m-1 \end{array}\right] \xrightarrow{R_3-2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & k-5 & m-3 \end{array}\right].$$

- $k \neq 5$: three leading entries in the coefficient part, so a unique solution.
- $k = 5$, $m \neq 3$: the last row reads $0 = m - 3 \neq 0$; inconsistent.
- $k = 5$, $m = 3$: the last row is zero, x_3 is free; infinitely many solutions.

Pitfall

Do not perform two operations "at once" that use a row being changed in the same step (e.g. $R_1 - R_2$ and $R_2 - R_1$ simultaneously); this can destroy information. Also, $R_i \to cR_i + R_j$ is fine only if $c \neq 0$; "$R_i \to 0R_i + R_j$" is not a row operation.

3.5 Rank and the structure of solution sets

Theorem 3.11 (System-Rank Theorem)**.** *Let A be $m \times n$.*

1. *$[A \mid \vec{b}]$ is inconsistent iff its RREF has a row $[0 \ \cdots \ 0 \mid 1]$, i.e. iff* $\operatorname{rank} A < \operatorname{rank}[A \mid \vec{b}]$.
2. *If the system is consistent, the solution set has $n - \operatorname{rank} A$ free parameters.*
3. *$A\vec{x} = \vec{b}$ is consistent for* every $\vec{b} \in \mathbb{R}^m$ *iff* $\operatorname{rank} A = m$.

Proof sketch. (1) and (2) are read off the RREF: each non-pivot column of A gives one free variable. For (3): if $\operatorname{rank} A = m$, every row of the RREF of A has a leading one, so a bad row $[0 \cdots 0 \mid 1]$ cannot occur. If $\operatorname{rank} A < m$, the RREF R of A has a zero row; start from $[R \mid \vec{e}_m]$ and reverse the row operations to get a $\vec{b}$ for which the system is inconsistent. □

Corollary 3.12. *Since* $\operatorname{rank} A \le \min(m,n)$*: a set of* k *vectors in* $\mathbb{R}^n$ *with* $k > n$ *is linearly dependent, and a set of* $k < n$ *vectors cannot span* $\mathbb{R}^n$*. Hence every basis of* $\mathbb{R}^n$ *has exactly* n *vectors.*

Proof. Put the vectors as columns of an $n \times k$ matrix A. Independence means $A\vec{c} = \vec{0}$ has only the trivial solution, i.e. $k - \operatorname{rank} A = 0$; but $\operatorname{rank} A \le n < k$. Spanning means $A\vec{c} = \vec{b}$ is consistent for all $\vec{b}$, i.e. $\operatorname{rank} A = n$; but $\operatorname{rank} A \le k < n$. □

Using row reduction for span and independence

For $\vec{v}_1, \dots, \vec{v}_k \in \mathbb{R}^n$ and $A = [\vec{v}_1 \ \cdots \ \vec{v}_k]$:

- $\vec{b} \in \operatorname{Span}\{\vec{v}_i\}$ iff $[A \mid \vec{b}]$ is consistent;
- $\{\vec{v}_i\}$ spans $\mathbb{R}^n$ iff $\operatorname{rank} A = n$;
- $\{\vec{v}_i\}$ is linearly independent iff $\operatorname{rank} A = k$.

3.6 Homogeneous systems

Definition 3.13. A system is *homogeneous* if all right-hand sides are 0: $A\vec{x} = \vec{0}$. It is always consistent, since $\vec{x} = \vec{0}$ (the *trivial solution*) works.

Theorem 3.14. *The solution set of a homogeneous system in* n *variables is a subspace of* $\mathbb{R}^n$ *(the* solution space*). Moreover, if* $\vec{x}_p$ *is any one solution of* $A\vec{x} = \vec{b}$*, then the full solution set of* $A\vec{x} = \vec{b}$ *is* $\{\vec{x}_p + \vec{x}_h : A\vec{x}_h = \vec{0}\}$.

Proof. If $\vec{x}, \vec{y}$ solve the homogeneous system then each equation gives $\sum_j a_{ij}(x_j + y_j) = 0 + 0$ and $\sum_j a_{ij}(cx_j) = c \cdot 0$. For the second claim, if $A\vec{x} = \vec{b}$ then $\vec{x} - \vec{x}_p$ solves the homogeneous system, and conversely $A(\vec{x}_p + \vec{x}_h) = \vec{b} + \vec{0}$. □

In the infinitely-many-solutions example above, $\vec{x}_p = (3,0,2,0)$ and the homogeneous solution space is $\operatorname{Span}\{(-2,1,0,0),(-1,0,-2,1)\}$; the spanning vectors produced by the free variables are automatically linearly independent (look at the free-variable positions).

Example 3.15. Find a basis for the solution space of $x_1 - x_2 + 2x_3 = 0$, $2x_1 - x_2 + x_3 = 0$.

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 2 & -1 & 1 & 0 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 1 & -3 & 0 \end{array}\right] \xrightarrow{R_1 + R_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -3 & 0 \end{array}\right].$$

With $x_3 = t$: $\vec{x} = t(1,3,1)$, so $\{(1,3,1)\}$ is a basis. Check: $1 - 3 + 2 = 0$, $2 - 3 + 1 = 0$.

Summary

- A linear system has 0, 1 or infinitely many solutions.
- Elementary row operations preserve the solution set; reduce to REF (Gaussian) or RREF (Gauss–Jordan). The RREF is unique; $\operatorname{rank} A$ = number of leading ones.
- Inconsistent iff a row $[0 \cdots 0 \mid 1]$ appears; otherwise $n - \operatorname{rank} A$ parameters.
- $A\vec{x} = \vec{b}$ is solvable for all $\vec{b}$ iff $\operatorname{rank} A = m$; columns independent iff $\operatorname{rank} A = n$.
- Homogeneous solution sets are subspaces; general solution = particular + homogeneous.

Chapter 4

Matrices

4.1 Matrix addition and scalar multiplication

Definition 4.1. An $m \times n$ *matrix* A is a rectangular array with m rows and n columns; its (i,j)-entry is written a_{ij} or $(A)_{ij}$. The set of all $m \times n$ real matrices is $M_{m\times n}(\mathbb{R})$. Two matrices are equal if they have the same size and the same entries. Addition and scalar multiplication are entrywise: $(A+B)_{ij} = a_{ij} + b_{ij}$, $(cA)_{ij} = ca_{ij}$. The *zero matrix* $O_{m,n}$ has all entries 0.

Theorem 4.2. *$M_{m\times n}(\mathbb{R})$ with these operations satisfies the same ten properties V1–V10 as $\mathbb{R}^n$ (with $O_{m,n}$ playing the role of $\vec{0}$).*

Definition 4.3 (Transpose)**.** The *transpose* of an $m \times n$ matrix A is the $n \times m$ matrix A^T with $(A^T)_{ij} = a_{ji}$: rows become columns. A is *symmetric* if $A^T = A$.

Theorem 4.4. $(A^T)^T = A$, $(A+B)^T = A^T + B^T$, $(cA)^T = cA^T$.

4.2 Matrix–vector multiplication

We write $A = [\vec{a}_1 \ \cdots \ \vec{a}_n]$ in terms of its columns $\vec{a}_j \in \mathbb{R}^m$.

Definition 4.5 (Matrix–vector product)**.** For A an $m \times n$ matrix and $\vec{x} \in \mathbb{R}^n$,

$$A\vec{x} = x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n \in \mathbb{R}^m.$$

So $A\vec{x}$ is a linear combination of the columns of A with weights from $\vec{x}$. Equivalently, if $\vec{r}_1^{\,T}, \ldots, \vec{r}_m^{\,T}$ are the rows of A, then the i-th entry of $A\vec{x}$ is the dot product $\vec{r}_i \cdot \vec{x}$. The system of Chapter 3 is now compactly $A\vec{x} = \vec{b}$.

Example 4.6.

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} = 2\begin{bmatrix} 1 \\ -1 \end{bmatrix} + 1\begin{bmatrix} 2 \\ 3 \end{bmatrix} - 1\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}.$$

Row view: $(1,2,0)\cdot(2,1,-1) = 4$ and $(-1,3,1)\cdot(2,1,-1) = -2+3-1 = 0$.

Theorem 4.7 (Linearity)**.** *$A(\vec{x}+\vec{y}) = A\vec{x} + A\vec{y}$ and $A(c\vec{x}) = c(A\vec{x})$. Also $A\vec{e}_j = \vec{a}_j$.*

Theorem 4.8 (Matrices Equal Theorem)**.** *If A and B are $m \times n$ and $A\vec{x} = B\vec{x}$ for every $\vec{x} \in \mathbb{R}^n$, then $A = B$.*

Proof. Take $\vec{x} = \vec{e}_j$: the j-th columns agree, $\vec{a}_j = A\vec{e}_j = B\vec{e}_j = \vec{b}_j$. □

4.3 Matrix multiplication

Definition 4.9 (Matrix product)**.** For A $m \times n$ and $B = [\vec{b}_1 \ \cdots \ \vec{b}_p]$ $n \times p$, the product is the $m \times p$ matrix

$$AB = [A\vec{b}_1 \ \ A\vec{b}_2 \ \cdots \ A\vec{b}_p], \qquad (AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B).$$

The product is defined only when the number of columns of A equals the number of rows of B.

The definition is chosen so that $(AB)\vec{x} = A(B\vec{x})$: multiplying by AB is the same as first multiplying by B then by A (composition; Chapter 5).

Example 4.10. Let $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 4 & 3 \end{bmatrix}$. Then

$$AB = \begin{bmatrix} 1\cdot 2 + 2\cdot 0 + 0\cdot 4 & 1\cdot 1 + 2(-1) + 0\cdot 3 \\ -1\cdot 2 + 3\cdot 0 + 1\cdot 4 & -1\cdot 1 + 3(-1) + 1\cdot 3 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 2 & -1 \end{bmatrix},$$

while BA is 3×3:

$$BA = \begin{bmatrix} 2-1 & 4+3 & 0+1 \\ 0+1 & 0-3 & 0-1 \\ 4-3 & 8+9 & 0+3 \end{bmatrix} = \begin{bmatrix} 1 & 7 & 1 \\ 1 & -3 & -1 \\ 1 & 17 & 3 \end{bmatrix}.$$

Theorem 4.11 (Properties of matrix multiplication)**.** *Whenever the sizes make sense, and $c \in \mathbb{R}$:*

1. $A(B+C) = AB + AC$ *and* $(A+B)C = AC + BC$*;*
2. $c(AB) = (cA)B = A(cB)$*;*
3. $A(BC) = (AB)C$ *(associativity);*
4. $(AB)^T = B^T A^T$*.*

Proof of (3) and (4). (3): $\big(A(BC)\big)_{ij} = \sum_k a_{ik} \sum_l b_{kl} c_{lj} = \sum_l \big(\sum_k a_{ik} b_{kl}\big) c_{lj} = \big((AB)C\big)_{ij}$. (4): $((AB)^T)_{ij} = (AB)_{ji} = \sum_k a_{jk} b_{ki} = \sum_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}$. □

What does *not* hold

- **Not commutative.** Even for square matrices, usually $AB \neq BA$:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

  Hence $(A+B)^2 = A^2 + AB + BA + B^2$, *not* $A^2 + 2AB + B^2$ in general.

- **No cancellation, zero divisors.** $AB = O$ does not imply $A = O$ or $B = O$:

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

  Similarly $AB = AC$ does not imply $B = C$.

- The order reverses in $(AB)^T = B^T A^T$.

4.4 The identity matrix and special matrices

Definition 4.12. The $n \times n$ *identity matrix* is $I_n = [\vec{e}_1 \ \cdots \ \vec{e}_n]$, with 1s on the diagonal and 0 elsewhere. A square matrix is *upper triangular* if all entries below the diagonal are 0, *lower triangular* if all entries above are 0, and *diagonal* if both; we write $\mathrm{diag}(d_1, \ldots, d_n)$.

Theorem 4.13. *For A $m \times n$, $I_m A = A = A I_n$. Moreover I_n is the* unique *matrix with $I\vec{x} = \vec{x}$ for all $\vec{x}$.*

Proof. $AI_n = [A\vec{e}_1 \ \cdots \ A\vec{e}_n] = [\vec{a}_1 \ \cdots \ \vec{a}_n] = A$. For $I_m A$, each column is $I_m \vec{a}_j = \sum_i (\vec{a}_j)_i \vec{e}_i = \vec{a}_j$. Uniqueness follows from the Matrices Equal Theorem. □

Powers of a square matrix are defined by $A^0 = I$, $A^{k+1} = A^k A$. Products of diagonal matrices are easy: $\mathrm{diag}(a_1, \ldots, a_n)\,\mathrm{diag}(b_1, \ldots, b_n) = \mathrm{diag}(a_1 b_1, \ldots, a_n b_n)$, so $\mathrm{diag}(d_1, \ldots, d_n)^k = \mathrm{diag}(d_1^k, \ldots, d_n^k)$. This is the key to Chapter 10.

4.5 Block multiplication

If matrices are partitioned into blocks of compatible sizes, they multiply "as if the blocks were numbers" (keeping the order of factors). For example, if $A = [A_1 \ A_2]$ (split by columns) and $B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$ (split by rows correspondingly), then $AB = A_1 B_1 + A_2 B_2$. A useful special case: the product of the column vector $\vec{u} \in \mathbb{R}^m$ with the row $\vec{v}^T$ ($\vec{v} \in \mathbb{R}^n$) is the $m \times n$ *outer product* $\vec{u}\vec{v}^T$, and

$$AB = \sum_{k=1}^{n} \vec{a}_k \vec{b}_k^T \quad (\vec{a}_k = \text{column } k \text{ of } A,\ \vec{b}_k^T = \text{row } k \text{ of } B).$$

Note also that $\vec{x} \cdot \vec{y} = \vec{x}^T \vec{y}$ (a 1×1 matrix), which gives the useful identity $(A\vec{x}) \cdot \vec{y} = \vec{x} \cdot (A^T \vec{y})$.

Proof of the identity. $(A\vec{x}) \cdot \vec{y} = (A\vec{x})^T \vec{y} = \vec{x}^T A^T \vec{y} = \vec{x} \cdot (A^T \vec{y})$. □

Summary

- $A\vec{x} = x_1 \vec{a}_1 + \cdots + x_n \vec{a}_n$; entry i is (row i)$\cdot\vec{x}$.
- $AB = [A\vec{b}_1 \cdots A\vec{b}_p]$, $(AB)_{ij} = \sum_k a_{ik} b_{kj}$; defined when inner sizes match.
- Associative and distributive, $(AB)^T = B^T A^T$; *not* commutative and no cancellation.
- $A\vec{x} = B\vec{x}$ for all $\vec{x}$ implies $A = B$. I is the multiplicative identity.

Chapter 5

Linear Mappings

5.1 Matrix mappings

Definition 5.1 (Matrix mapping)**.** For an $m \times n$ matrix A, the function $f_A : \mathbb{R}^n \to \mathbb{R}^m$ defined by $f_A(\vec{x}) = A\vec{x}$ is the *matrix mapping* corresponding to A. $\mathbb{R}^n$ is the *domain* and $\mathbb{R}^m$ the *codomain*.

By linearity of matrix–vector multiplication, every matrix mapping satisfies $f_A(b\vec{x} + c\vec{y}) = bf_A(\vec{x}) + cf_A(\vec{y})$. This property is what we abstract.

Definition 5.2 (Linear mapping)**.** A function $L : \mathbb{R}^n \to \mathbb{R}^m$ is a *linear mapping* if for all $\vec{x}, \vec{y} \in \mathbb{R}^n$ and $b, c \in \mathbb{R}$,

$$L(b\vec{x} + c\vec{y}) = bL(\vec{x}) + cL(\vec{y}).$$

A linear mapping $L : \mathbb{R}^n \to \mathbb{R}^n$ is called a *linear operator*.

Immediate consequences: $L(\vec{0}) = \vec{0}$ (take $b = c = 0$), and $L(c_1\vec{v}_1 + \cdots + c_k\vec{v}_k) = c_1L(\vec{v}_1) + \cdots + c_kL(\vec{v}_k)$.

Example 5.3. $L(x_1, x_2, x_3) = (x_1 + 2x_2,\ 3x_2 - x_3)$ is linear: for $\vec{x}, \vec{y}$ and b, c,

$$L(b\vec{x} + c\vec{y}) = (bx_1 + cy_1 + 2bx_2 + 2cy_2,\ 3bx_2 + 3cy_2 - bx_3 - cy_3) = bL(\vec{x}) + cL(\vec{y}).$$

In contrast, $M(x_1, x_2) = (x_1+1, x_2)$ is not linear since $M(\vec{0}) = (1, 0) \neq \vec{0}$, and $N(x_1, x_2) = (x_1x_2, 0)$ is not linear since $N(2(1, 1)) = (4, 0) \neq 2N(1, 1) = (2, 0)$.

5.2 The standard matrix

Theorem 5.4. *Every linear mapping $L : \mathbb{R}^n \to \mathbb{R}^m$ is a matrix mapping: $L(\vec{x}) = [L]\vec{x}$ where*

$$[L] = \begin{bmatrix} L(\vec{e}_1) & L(\vec{e}_2) & \cdots & L(\vec{e}_n) \end{bmatrix}$$

is the standard matrix *of L. It is the unique matrix with this property.*

Proof. $L(\vec{x}) = L(x_1\vec{e}_1 + \cdots + x_n\vec{e}_n) = x_1L(\vec{e}_1) + \cdots + x_nL(\vec{e}_n) = [L]\vec{x}$ by the definition of matrix–vector multiplication. Uniqueness is the Matrices Equal Theorem. □

Example 5.5. For $L(x_1, x_2, x_3) = (x_1 + 2x_2,\ 3x_2 - x_3)$: $L(\vec{e}_1) = (1, 0)$, $L(\vec{e}_2) = (2, 3)$, $L(\vec{e}_3) = (0, -1)$, so $[L] = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & -1 \end{bmatrix}$.

So "linear mapping $\mathbb{R}^n \to \mathbb{R}^m$" and "$m \times n$ matrix" are two views of the same object.

5.3 Geometric transformations

Rotations

$R_\theta : \mathbb{R}^2 \to \mathbb{R}^2$ rotates counter-clockwise by θ. Rotation sends $\vec{e}_1$ to $(\cos\theta, \sin\theta)$ and $\vec{e}_2$ to $(-\sin\theta, \cos\theta)$, and (geometrically) preserves sums and scalings, so it is linear with

$$[R_\theta] = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

For example $[R_{\pi/2}] = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

Projections

For $\vec{d} \neq \vec{0}$, $\text{proj}_{\vec{d}}$ is linear (Chapter 2). For $\vec{d} = (1,2)$, $\text{proj}_{\vec{d}}\vec{e}_1 = \frac{1}{5}(1,2)$ and $\text{proj}_{\vec{d}}\vec{e}_2 = \frac{2}{5}(1,2)$, so

$$[\text{proj}_{\vec{d}}] = \frac{1}{5}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

In general $[\text{proj}_{\vec{d}}] = \dfrac{1}{\left\|\vec{d}\right\|^2}\vec{d}\vec{d}^T$.

Reflections

Reflecting $\vec{x}$ in the line $\text{Span}\{\vec{d}\}$ keeps the projection and flips the perpendicular part: $\text{refl}_{\vec{d}}\vec{x} = \text{proj}_{\vec{d}}\vec{x} - \text{perp}_{\vec{d}}\vec{x} = 2\,\text{proj}_{\vec{d}}\vec{x} - \vec{x}$. For $\vec{d} = (1,2)$,

$$[\text{refl}_{\vec{d}}] = \frac{2}{5}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} - I = \frac{1}{5}\begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix}.$$

Check: $\frac{1}{5}(-3+8,\ 4+6) = (1,2)$, so $\vec{d}$ is fixed, and $(2,-1) \mapsto \frac{1}{5}(-6-4,\ 8-3) = (-2,1)$ is flipped. A reflection in a plane through the origin in $\mathbb{R}^3$ with normal $\vec{n}$ is $\text{refl}_P\,\vec{x} = \vec{x} - 2\,\text{proj}_{\vec{n}}\,\vec{x}$.

Stretches and shears

$\begin{bmatrix} t & 0 \\ 0 & 1 \end{bmatrix}$ stretches horizontally by t; $\begin{bmatrix} 1 & s \\ 0 & 1 \end{bmatrix}$ is a horizontal shear, $(x_1, x_2) \mapsto (x_1 + sx_2, x_2)$.

5.4 Operations on linear mappings and composition

Definition 5.6. For $L, M : \mathbb{R}^n \to \mathbb{R}^m$ and $c \in \mathbb{R}$, define $(L+M)(\vec{x}) = L(\vec{x}) + M(\vec{x})$ and $(cL)(\vec{x}) = cL(\vec{x})$. For $M : \mathbb{R}^n \to \mathbb{R}^m$ and $L : \mathbb{R}^m \to \mathbb{R}^p$, the *composition* is $(L \circ M)(\vec{x}) = L(M(\vec{x}))$.

Theorem 5.7. *These are linear, and* $[L+M] = [L] + [M]$, $[cL] = c[L]$, $[L \circ M] = [L][M]$.

Proof for composition. $(L \circ M)(\vec{x}) = [L]([M]\vec{x}) = ([L][M])\vec{x}$ by associativity, so $L \circ M$ is the matrix mapping of $[L][M]$, which is linear. □

Example 5.8. Rotate by $\pi/2$ and then reflect in the x_1-axis ($F = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$):

$$[F \circ R_{\pi/2}] = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}0 & -1\\ -1 & 0\end{bmatrix},$$

which is the reflection in the line $x_2 = -x_1$. In the other order, $[R_{\pi/2}][F] = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$, the reflection in $x_2 = x_1$; order matters. Also $[R_\alpha][R_\beta] = [R_{\alpha+\beta}]$, which (multiplying out) re-proves the angle-sum formulas.

5.5 Range and kernel

Definition 5.9. For a linear mapping $L : \mathbb{R}^n \to \mathbb{R}^m$,

$$\text{Range}(L) = \{L(\vec{x}) : \vec{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, \qquad \text{Ker}(L) = \{\vec{x} \in \mathbb{R}^n : L(\vec{x}) = \vec{0}\} \subseteq \mathbb{R}^n.$$

Theorem 5.10. $\text{Range}(L)$ *is a subspace of* $\mathbb{R}^m$ *and* $\text{Ker}(L)$ *is a subspace of* $\mathbb{R}^n$. *If* $A = [L]$, $\text{Range}(L) = \text{Span}\{\vec{a}_1, \ldots, \vec{a}_n\}$ *(the column space of* A*) and* $\text{Ker}(L)$ *is the solution space of* $A\vec{x} = \vec{0}$.

Proof. $L(\vec{0}) = \vec{0}$ puts $\vec{0}$ in both. If $L(\vec{x}) = L(\vec{y}) = \vec{0}$ then $L(\vec{x} + c\vec{y}) = \vec{0} + c\vec{0} = \vec{0}$. If $\vec{u} = L(\vec{x}), \vec{v} = L(\vec{y})$ then $\vec{u} + c\vec{v} = L(\vec{x} + c\vec{y}) \in \text{Range}(L)$. The descriptions follow from $L(\vec{x}) = A\vec{x} = x_1\vec{a}_1 + \cdots + x_n\vec{a}_n$. □

Example 5.11. $L(\vec{x}) = A\vec{x}$ with $A = \begin{bmatrix}1 & 1 & 2\\ 1 & 2 & 3\end{bmatrix}$.

$$\begin{bmatrix}1 & 1 & 2\\ 1 & 2 & 3\end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix}1 & 1 & 2\\ 0 & 1 & 1\end{bmatrix} \xrightarrow{R_1 - R_2} \begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 1\end{bmatrix}.$$

Kernel: $x_3 = t$, $x_1 = x_2 = -t$, so $\text{Ker}(L) = \text{Span}\{(-1, -1, 1)\}$. Since $\text{rank}\, A = 2 = m$, $A\vec{x} = \vec{b}$ is consistent for all $\vec{b}$, so $\text{Range}(L) = \mathbb{R}^2$.

Example 5.12. $L(x_1, x_2) = (x_1 + x_2,\ 2x_1 + 2x_2,\ x_1 + x_2)$ has $[L] = \begin{bmatrix}1 & 1\\ 2 & 2\\ 1 & 1\end{bmatrix}$, so $\text{Range}(L) = \text{Span}\{(1, 2, 1)\}$ (a line in $\mathbb{R}^3$) and $\text{Ker}(L) = \text{Span}\{(1, -1)\}$.

Geometric kernels and ranges

For $\text{proj}_{\vec{d}}$ on $\mathbb{R}^2$: range = the line $\text{Span}\{\vec{d}\}$, kernel = the line orthogonal to $\vec{d}$. Rotations and reflections have kernel $\{\vec{0}\}$ and range all of $\mathbb{R}^2$.

Theorem 5.13. L *is one-to-one iff* $\text{Ker}(L) = \{\vec{0}\}$. $L : \mathbb{R}^n \to \mathbb{R}^m$ *is onto iff* $\text{rank}[L] = m$.

Proof of the first claim. If L is one-to-one and $L(\vec{x}) = \vec{0} = L(\vec{0})$ then $\vec{x} = \vec{0}$. Conversely if the kernel is trivial and $L(\vec{x}) = L(\vec{y})$, then $L(\vec{x} - \vec{y}) = \vec{0}$ so $\vec{x} = \vec{y}$. □

Summary

- Linear: $L(b\vec{x} + c\vec{y}) = bL(\vec{x}) + cL(\vec{y})$; then $L(\vec{0}) = \vec{0}$.
- Every linear $L : \mathbb{R}^n \to \mathbb{R}^m$ is $\vec{x} \mapsto [L]\vec{x}$ with $[L] = [L(\vec{e}_1) \cdots L(\vec{e}_n)]$.
- $[R_\theta]$, $[\text{proj}_{\vec{d}}] = \vec{d}\vec{d}^T / \left\|\vec{d}\right\|^2$, $[\text{refl}_{\vec{d}}] = 2[\text{proj}_{\vec{d}}] - I$.
- $[L \circ M] = [L][M]$: composition is matrix multiplication (order matters).
- Range(L) = span of the columns of $[L]$; Ker(L) = solutions of $[L]\vec{x} = \vec{0}$; both are subspaces. One-to-one iff trivial kernel.

Chapter 6

The Four Fundamental Subspaces

6.1 Definitions

Definition 6.1. Let A be an $m \times n$ matrix. The four *fundamental subspaces* of A are

$$\begin{aligned} \mathrm{Col}(A) &= \{A\vec{x} : \vec{x} \in \mathbb{R}^n\} = \mathrm{Span}\{\text{columns of } A\} \subseteq \mathbb{R}^m && \text{(column space)} \\ \mathrm{Row}(A) &= \mathrm{Col}(A^T) = \mathrm{Span}\{\text{rows of } A\} \subseteq \mathbb{R}^n && \text{(row space)} \\ \mathrm{Null}(A) &= \{\vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0}\} \subseteq \mathbb{R}^n && \text{(null space)} \\ \mathrm{Null}(A^T) &= \{\vec{y} \in \mathbb{R}^m : A^T\vec{y} = \vec{0}\} \subseteq \mathbb{R}^m && \text{(left null space)} \end{aligned}$$

If $L(\vec{x}) = A\vec{x}$ then $\mathrm{Col}(A) = \mathrm{Range}(L)$ and $\mathrm{Null}(A) = \mathrm{Ker}(L)$. All four are subspaces (they are spans or solution spaces of homogeneous systems).

6.2 Dimension of a subspace of $\mathbb{R}^n$

Theorem 6.2. *If S is a non-trivial subspace of $\mathbb{R}^n$, then S has a basis, and every basis of S has the same number of vectors.*

Proof sketch. Start with a non-zero vector of S and keep adding vectors of S not in the current span; each addition keeps the set independent (Chapter 1), and by Chapter 3 an independent set in $\mathbb{R}^n$ has at most n vectors, so the process stops with a basis. If $\{\vec{v}_1, \ldots, \vec{v}_k\}$ spans S and $\{\vec{w}_1, \ldots, \vec{w}_\ell\} \subseteq S$ with $\ell > k$, write each $\vec{w}_j$ in terms of the $\vec{v}_i$: $W = VC$ with C a $k \times \ell$ matrix. Since $\ell > k$ there is $\vec{c} \neq \vec{0}$ with $C\vec{c} = \vec{0}$, so $W\vec{c} = \vec{0}$ and the $\vec{w}_j$ are dependent. Applying this both ways shows two bases have the same size. □

Definition 6.3 (Dimension)**.** The *dimension* $\dim S$ of a subspace S is the number of vectors in any basis; $\dim\{\vec{0}\} = 0$.

Theorem 6.4. *If $\dim S = k$, then any k linearly independent vectors in S form a basis, and any k vectors that span S form a basis. If $S \subseteq T$ are subspaces then $\dim S \leq \dim T$, with equality iff $S = T$.*

6.3 Finding bases

The key observation: row operations change the columns of A, but they preserve every linear relation among the columns (since $A\vec{c} = \vec{0}$ iff $R\vec{c} = \vec{0}$ when A and R are row equivalent). They also do not change the row space (new rows are combinations of old rows and vice versa).

Theorem 6.5. *Let R be the RREF of A (or any REF).*

1. *The pivot columns* of A *(not of R) form a basis for* $\mathrm{Col}(A)$.
2. *The non-zero rows of R form a basis for* $\mathrm{Row}(A)$.
3. *The vectors obtained from the parametric vector form of the solution to $A\vec{x} = \vec{0}$ (one per free variable) form a basis for* $\mathrm{Null}(A)$.

Consequently $\dim \mathrm{Col}(A) = \dim \mathrm{Row}(A) = \operatorname{rank} A$ *and* $\dim \mathrm{Null}(A) = n - \operatorname{rank} A$.

Proof sketch. (1) In R the pivot columns are distinct standard basis vectors, hence independent, and every non-pivot column is a combination of them. These relations transfer to the columns of A. (2) The non-zero rows of R span $\mathrm{Row}(R) = \mathrm{Row}(A)$ and are independent because each has a leading one in a column where the others have zeros. (3) Each basic vector has a 1 in its own free variable's position and 0 in the other free positions, giving independence; spanning is the parametric form itself. □

Pitfall

Row operations *change* the column space. The columns of R tell you *which* columns to take, but the basis vectors must come from A itself. Conversely, for the row space you may use the rows of R (and the rows of A in the pivot positions are *not* necessarily a basis if you swapped rows).

Example 6.6. Let $A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 4 \\ 1 & 2 & 1 & 3 \end{bmatrix}$. As in Chapter 3,

$$A \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R.$$

$\operatorname{rank} A = 2$, with pivots in columns 1 and 3.

- $\mathrm{Col}(A)$: basis $\{(1,2,1),(0,1,1)\}$ (columns 1 and 3 of A), dimension 2.
- $\mathrm{Row}(A)$: basis $\{(1,2,0,1),(0,0,1,2)\}$, dimension 2.
- $\mathrm{Null}(A)$: $x_2 = s$, $x_4 = t$, $x_1 = -2s - t$, $x_3 = -2t$, so the basis is $\{(-2,1,0,0),(-1,0,-2,1)\}$, dimension $2 = 4 - 2$.
- $\mathrm{Null}(A^T)$: row reduce A^T:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \\ 1 & 4 & 3 \end{bmatrix} \xrightarrow[R_4 - R_1]{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix} \xrightarrow[R_2 \leftrightarrow R_3]{R_4 - 2R_3} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 - 2R_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

  So $y_3 = t$, $y_1 = t$, $y_2 = -t$: basis $\{(1,-1,1)\}$, dimension $1 = 3 - 2$.

Sanity check: $(1,-1,1)$ is orthogonal to both basis vectors of $\mathrm{Col}(A)$: $1-2+1 = 0$ and $0-1+1 = 0$. Likewise $(1,2,0,1)\cdot(-2,1,0,0) = 0$, $(1,2,0,1)\cdot(-1,0,-2,1) = 0$, $(0,0,1,2)\cdot(-2,1,0,0) = 0$, $(0,0,1,2)\cdot(-1,0,-2,1) = 0$.

6.4 The Rank–Nullity Theorem

Definition 6.7. The *nullity* of A is $\operatorname{nullity} A = \dim \operatorname{Null}(A)$.

Theorem 6.8 (Rank–Nullity / Dimension Theorem)**.** *For any $m \times n$ matrix A,*

$$\operatorname{rank} A + \operatorname{nullity} A = n.$$

Equivalently, for linear $L : \mathbb{R}^n \to \mathbb{R}^m$, $\dim \operatorname{Range}(L) + \dim \operatorname{Ker}(L) = n$.

Proof. In the RREF, each of the n columns is either a pivot column (there are $\operatorname{rank} A$ of them) or corresponds to a free variable (each contributing one basis vector of $\operatorname{Null}(A)$). □

Theorem 6.9. $\operatorname{rank} A = \operatorname{rank} A^T$. *Hence* $\dim \operatorname{Null}(A^T) = m - \operatorname{rank} A$.

Proof. $\operatorname{rank} A^T = \dim \operatorname{Col}(A^T) = \dim \operatorname{Row}(A) = \operatorname{rank} A$. □

Example 6.10. A 5×7 matrix A with $\operatorname{rank} A = 3$ has $\dim \operatorname{Col} A = \dim \operatorname{Row} A = 3$, $\operatorname{nullity} A = 7 - 3 = 4$ and $\dim \operatorname{Null}(A^T) = 5 - 3 = 2$. It cannot have rank 6, since $\operatorname{rank} A \le \min(5, 7)$.

6.5 Orthogonality of the fundamental subspaces

Definition 6.11. For a subspace S of $\mathbb{R}^n$, its *orthogonal complement* is $S^\perp = \{\vec{x} \in \mathbb{R}^n : \vec{x} \cdot \vec{s} = 0 \text{ for all } \vec{s} \in S\}$.

Theorem 6.12. $\operatorname{Null}(A) = \operatorname{Row}(A)^\perp$ *in* $\mathbb{R}^n$, *and* $\operatorname{Null}(A^T) = \operatorname{Col}(A)^\perp$ *in* $\mathbb{R}^m$.

Proof. $A\vec{x} = \vec{0}$ says exactly that $\vec{x}$ has dot product zero with every row of A, hence with every linear combination of rows. Apply this to A^T for the second statement. □

Together with rank–nullity this says $\mathbb{R}^n$ is split into two orthogonal pieces of dimensions $\operatorname{rank} A$ and $n - \operatorname{rank} A$, and similarly $\mathbb{R}^m$. This also gives a test for consistency: $A\vec{x} = \vec{b}$ is consistent iff $\vec{b} \in \operatorname{Col}(A)$ iff $\vec{b}$ is orthogonal to every vector in $\operatorname{Null}(A^T)$. In the example above, $A\vec{x} = \vec{b}$ is consistent iff $b_1 - b_2 + b_3 = 0$.

Summary

- $\operatorname{Col} A$, $\operatorname{Row} A$, $\operatorname{Null} A$, $\operatorname{Null} A^T$; $\operatorname{Col} A = \operatorname{Range}$, $\operatorname{Null} A = \operatorname{Ker}$.
- Basis of $\operatorname{Col} A$: pivot columns of A. Basis of $\operatorname{Row} A$: non-zero rows of the RREF. Basis of $\operatorname{Null} A$: vectors from the free variables.
- $\dim \operatorname{Col} A = \dim \operatorname{Row} A = \operatorname{rank} A$; rank–nullity: $\operatorname{rank} A + \operatorname{nullity} A = n$.
- $\operatorname{Null} A \perp \operatorname{Row} A$ and $\operatorname{Null} A^T \perp \operatorname{Col} A$.

Chapter 7

Inverse Matrices and Elementary Matrices

7.1 Inverses

Definition 7.1. An $n \times n$ matrix A is *invertible* if there is an $n \times n$ matrix B with $AB = I = BA$. Then B is *the inverse* of A, written A^{-1}. A non-invertible square matrix is called *singular*.

The inverse is unique: if B and C are both inverses then $B = BI = B(AC) = (BA)C = IC = C$. Only square matrices have (two-sided) inverses.

Theorem 7.2 (One-sided suffices)**.** *If A, B are $n \times n$ and $AB = I$, then $BA = I$; so A and B are inverses of each other.*

Proof. If $B\vec{x} = \vec{0}$ then $\vec{x} = I\vec{x} = A(B\vec{x}) = \vec{0}$, so $\text{Null}(B) = \{\vec{0}\}$ and $\operatorname{rank} B = n$. Hence $B\vec{x} = \vec{y}$ is solvable for every $\vec{y}$. For any $\vec{y}$ write $\vec{y} = B\vec{x}$; then $(BA)\vec{y} = BAB\vec{x} = B(AB)\vec{x} = B\vec{x} = \vec{y}$. So $BA\vec{y} = I\vec{y}$ for all $\vec{y}$, and $BA = I$ by the Matrices Equal Theorem. □

Theorem 7.3 (Properties)**.** *If A, B are invertible $n \times n$ and $c \neq 0$:*

1. $(A^{-1})^{-1} = A$ *and* $(cA)^{-1} = \frac{1}{c}A^{-1}$;
2. $(AB)^{-1} = B^{-1}A^{-1}$ *(note the order);*
3. $(A^T)^{-1} = (A^{-1})^T$;
4. $(A^k)^{-1} = (A^{-1})^k$.

Proof of (2) and (3). $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$. And $A^T(A^{-1})^T = (A^{-1}A)^T = I^T = I$. □

Theorem 7.4 (2×2 inverse)**.** $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ *is invertible iff $ad - bc \neq 0$, and then*

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Proof. Multiply out: $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = (ad - bc)I$. If $ad - bc = 0$, this product shows a non-zero vector (a column of the second matrix, unless $A = O$) in $\text{Null}(A)$, so A is not invertible. □

7.2 Computing inverses by row reduction

To find A^{-1} we must solve $A\vec{b}_j = \vec{e}_j$ for each column $\vec{b}_j$ of $B = A^{-1}$. All n systems have the same coefficient matrix, so we solve them at once.

Matrix Inversion Algorithm

Row reduce $[A \mid I]$. If the RREF has the form $[I \mid B]$ then A is invertible and $B = A^{-1}$. If the left block does not reduce to I (i.e. $\operatorname{rank} A < n$), then A is not invertible.

Example 7.5. Find the inverse of $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3-R_1]{R_2-2R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3+2R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right] \xrightarrow{-R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$$\xrightarrow[R_2+3R_3]{R_1-3R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right] \xrightarrow{R_1-2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right].$$

So $A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$. Check the first row of AA^{-1}: $(1,2,3)\cdot(-40,13,5) = -40+26+15 = 1$, $(1,2,3)\cdot(16,-5,-2) = 16-10-6 = 0$, $(1,2,3)\cdot(9,-3,-1) = 9-6-3 = 0$.

Example 7.6 (A singular matrix)**.**

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right].$$

The left block has rank $1 < 2$, so $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is not invertible (consistent with $ad - bc = 4 - 4 = 0$).

If A is invertible, then $A\vec{x} = \vec{b}$ has the unique solution $\vec{x} = A^{-1}\vec{b}$. (In practice row reducing $[A \mid \vec{b}]$ is cheaper than computing A^{-1}.)

7.3 Elementary matrices

Definition 7.7. An *elementary matrix* is a matrix obtained from I_n by performing exactly one elementary row operation.

Theorem 7.8. *If E is the elementary matrix for a row operation, then EA is the result of performing that operation on A. Every elementary matrix is invertible, and E^{-1} is the elementary matrix of the reverse operation.*

Example 7.9. In 3×3: $E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ performs $R_2 - 2R_1$, with $E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (performing $R_2 + 2R_1$).

Theorem 7.10. *If A row reduces to R via operations with elementary matrices $E_1, \ldots, E_k$, then $E_k \cdots E_2 E_1 A = R$. In particular if A is invertible then $R = I$, and*

$$A^{-1} = E_k \cdots E_1, \qquad A = E_1^{-1} E_2^{-1} \cdots E_k^{-1},$$

so every invertible matrix is a product of elementary matrices.

This also explains the inversion algorithm: applying $E_k \cdots E_1 = A^{-1}$ to $[A \mid I]$ gives $[I \mid A^{-1}]$.

Example 7.11. Write $A = \begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix}$ as a product of elementary matrices.

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix} \xrightarrow{-\frac{1}{2}R_2} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_1 - 3R_2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

With $E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $E_2 = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$, $E_3 = \begin{bmatrix} 1 & 0 \\ 0 & -\frac{1}{2} \end{bmatrix}$, $E_4 = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$ we get $E_4 E_3 E_2 E_1 A = I$, so

$$A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}.$$

Check: the last two give $\begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix}$; times $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ on the left gives $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$; swapping rows gives A.
Also $A^{-1} = E_4 E_3 E_2 E_1 = \frac{1}{2}\begin{bmatrix} 3 & -4 \\ -1 & 2 \end{bmatrix}$, matching the 2×2 formula with $ad - bc = 2$.

7.4 The Invertible Matrix Theorem

Theorem 7.12 (Invertible Matrix Theorem)**.** *For an $n \times n$ matrix A, the following are equivalent:*

1. *A is invertible;*
2. *the RREF of A is I;*
3. *$\operatorname{rank} A = n$;*
4. *$A\vec{x} = \vec{b}$ is consistent with a unique solution for every $\vec{b} \in \mathbb{R}^n$;*
5. *$A\vec{x} = \vec{0}$ has only the trivial solution, i.e. $\operatorname{Null}(A) = \{\vec{0}\}$;*
6. *the columns of A are linearly independent;*
7. *the columns of A span $\mathbb{R}^n$ (form a basis of $\mathbb{R}^n$);*
8. *the rows of A form a basis of $\mathbb{R}^n$;*
9. *A^T is invertible;*

10. *$L(\vec{x}) = A\vec{x}$ is one-to-one; equivalently, onto;*
11. *A is a product of elementary matrices;*
12. $\det A \neq 0$ *(Chapter 8);*
13. *0 is not an eigenvalue of A (Chapter 9).*

Proof sketch. (2)⇔(3) by definition of rank for square matrices. (3)⇔(4)⇔(5) by the System-Rank Theorem ($n - \operatorname{rank} A = 0$ parameters and $\operatorname{rank} A = n = m$). (5)⇔(6) and (3)⇔(7) are the row-reduction criteria for independence and spanning. (1)⇒(5): $A\vec{x} = \vec{0}$ gives $\vec{x} = A^{-1}\vec{0} = \vec{0}$. (2)⇒(11)⇒(1) by the previous section. (1)⇔(9) from $(A^T)^{-1} = (A^{-1})^T$, and (8) is (6) for A^T. □

Pitfalls

$(A + B)^{-1} \neq A^{-1} + B^{-1}$ in general (e.g. $A = B = I$ gives $(2I)^{-1} = \frac{1}{2}I \neq 2I$). And do not "divide" by matrices: from $AB = AC$ you may conclude $B = C$ *only* if A is invertible (multiply on the left by A^{-1}).

Summary

- A^{-1} satisfies $AA^{-1} = A^{-1}A = I$; one side suffices for square matrices; it is unique.
- $(AB)^{-1} = B^{-1}A^{-1}$, $(A^T)^{-1} = (A^{-1})^T$; 2×2 formula with $ad - bc$.
- Row reduce $[A \mid I]$ to $[I \mid A^{-1}]$.
- Elementary matrices perform row operations by left multiplication; invertible matrices are products of elementary matrices.
- Invertible Matrix Theorem: invertible ⇔ rank n ⇔ trivial null space ⇔ columns form a basis ⇔ $\det A \neq 0$.

Chapter 8

Determinants

8.1 Definition by cofactor expansion

For 2×2 matrices we already met $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad-bc$, which decides invertibility. We extend this recursively.

Definition 8.1 (Minor, cofactor, determinant)**.** Let A be $n\times n$ with $n\geq 2$. Let $A(i,j)$ be the $(n-1)\times(n-1)$ matrix obtained by deleting row i and column j of A. The *cofactor* of a_{ij} is

$$C_{ij} = (-1)^{i+j}\det A(i,j).$$

The *determinant* is $\det A = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}$ (expansion along the first row), with $\det[a] = a$ for 1×1 matrices. We also write $|A|$ for $\det A$.

The signs $(-1)^{i+j}$ follow a checkerboard pattern starting with $+$ in the top-left corner.

Theorem 8.2 (Cofactor expansion along any row or column)**.** *For any i and any j,*

$$\det A = \sum_{k=1}^{n} a_{ik}C_{ik} = \sum_{k=1}^{n} a_{kj}C_{kj}.$$

The proof is a (tedious) induction on n. Practically: expand along a row or column with many zeros.

Example 8.3. For $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$, along the first row:

$$\det A = 1\begin{vmatrix} 5 & 3 \\ 0 & 8 \end{vmatrix} - 2\begin{vmatrix} 2 & 3 \\ 1 & 8 \end{vmatrix} + 3\begin{vmatrix} 2 & 5 \\ 1 & 0 \end{vmatrix} = 1(40) - 2(16-3) + 3(0-5) = 40 - 26 - 15 = -1.$$

Along the second column (which has a zero): $\det A = -2\begin{vmatrix} 2 & 3 \\ 1 & 8 \end{vmatrix} + 5\begin{vmatrix} 1 & 3 \\ 1 & 8 \end{vmatrix} - 0 = -2(13)+5(5) = -1$. Same answer.

Theorem 8.4. *If A is upper or lower triangular, $\det A$ is the product of the diagonal entries. In particular $\det I = 1$. If A has a row or column of zeros, $\det A = 0$.*

Proof. For lower triangular, expand along the first row: only $a_{11}C_{11}$ survives, and $A(1,1)$ is again lower triangular; induct. (Upper triangular: expand along the first column.) □

8.2 Determinants and row operations

Theorem 8.5 (Effect of row operations)**.** *Let A be $n \times n$.*

1. *If B is obtained by multiplying one row of A by c, then* $\det B = c \det A$.
2. *If B is obtained by swapping two rows, then* $\det B = -\det A$.
3. *If B is obtained by adding a multiple of one row to another, then* $\det B = \det A$.

Proof sketch. (1) Expand along the scaled row. (2) Induction: for $n \geq 3$, expand along a row not involved in the swap; each minor has two rows swapped. The 2×2 case is direct. Consequently a matrix with two equal rows has determinant 0 (swapping them gives $\det A = -\det A$). (3) Expanding B along row i (where $R_i \to R_i + cR_j$) gives $\det A + c \det A'$, where A' has row j appearing twice, so $\det A' = 0$. □

Strategy

Row reduce to upper triangular form, tracking swaps (each flips the sign) and scalings (each multiplies by the scalar), then multiply the diagonal. Because $\det A^T = \det A$, column operations behave the same way.

Example 8.6. Compute $\det B$ for $B = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 3 \\ 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & 4 \end{bmatrix}$.

$$\begin{vmatrix} 1 & 2 & 0 & 1 \\ 2 & 5 & 1 & 3 \\ 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & 4 \end{vmatrix} \overset{\substack{R_2-2R_1 \\ R_4-R_1}}{=} \begin{vmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & 3 \end{vmatrix} \overset{R_3-R_2}{=} \begin{vmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 \end{vmatrix}$$

$$\overset{R_3 \leftrightarrow R_4}{=} - \begin{vmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 2 & 1 \end{vmatrix} \overset{R_4-2R_3}{=} - \begin{vmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & -5 \end{vmatrix}.$$

So $\det B = -(1 \cdot 1 \cdot 1 \cdot (-5)) = 5$.

Theorem 8.7. *A is invertible if and only if* $\det A \neq 0$.

Proof. Row reduce A to its RREF R; each operation multiplies the determinant by a non-zero number, so $\det A \neq 0$ iff $\det R \neq 0$. For square RREF, $R = I$ (determinant 1) if $\operatorname{rank} A = n$, and otherwise R has a zero row (determinant 0). □

8.3 Products and transposes

Theorem 8.8. *For $n \times n$ matrices A, B:*

1. $\det(EA) = \det E \det A$ *for any elementary matrix E;*
2. $\det(AB) = \det A \det B$*;*

3. $\det(A^T) = \det A$;

4. *if A is invertible,* $\det(A^{-1}) = \dfrac{1}{\det A}$;

5. $\det(cA) = c^n \det A$.

Proof sketch. (1) is the row-operation theorem, since $\det E$ is c, -1 or 1 respectively. (2) If A is invertible, write $A = E_1 \cdots E_k$ and apply (1) repeatedly. If not, AB is not invertible either (if AB had inverse C then $A(BC) = I$ would make A invertible), so both sides are 0. (4) follows from $\det A \det A^{-1} = \det I = 1$. (5): cA scales each of n rows by c. □

Pitfall

$\det(A + B) \neq \det A + \det B$ in general, and $\det(cA) = c^n \det A$, *not* $c \det A$. For example $\det(2I_3) = 8$.

8.4 The adjugate and a formula for the inverse

Definition 8.9. The *cofactor matrix* of A is $\operatorname{cof} A = [C_{ij}]$, and the *adjugate* is $\operatorname{adj} A = (\operatorname{cof} A)^T$, i.e. $(\operatorname{adj} A)_{ij} = C_{ji}$.

Theorem 8.10. $A \operatorname{adj} A = (\det A) I = \operatorname{adj} A \, A$. *Hence if* $\det A \neq 0$, $A^{-1} = \dfrac{1}{\det A} \operatorname{adj} A$.

Proof. $(A \operatorname{adj} A)_{ij} = \sum_k a_{ik} C_{jk}$. If $i = j$ this is the expansion of $\det A$ along row i. If $i \neq j$ it is the expansion along row j of the matrix obtained from A by replacing row j with row i; that matrix has two equal rows, so the sum is 0 ("expansion by alien cofactors"). □

Example 8.11. For $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$ (with $\det A = -1$), the cofactors are

$$\begin{aligned} &C_{11} = 40, && C_{12} = -(16 - 3) = -13, && C_{13} = 0 - 5 = -5, \\ &C_{21} = -(16 - 0) = -16, && C_{22} = 8 - 3 = 5, && C_{23} = -(0 - 2) = 2, \\ &C_{31} = 6 - 15 = -9, && C_{32} = -(3 - 6) = 3, && C_{33} = 5 - 4 = 1. \end{aligned}$$

So $\operatorname{adj} A = \begin{bmatrix} 40 & -16 & -9 \\ -13 & 5 & 3 \\ -5 & 2 & 1 \end{bmatrix}$ and $A^{-1} = \frac{1}{-1} \operatorname{adj} A = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$, agreeing with the row-reduction computation of Chapter 7.

For a 2×2 matrix the adjugate is $\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, recovering the familiar formula.

8.5 Cramer's Rule

Theorem 8.12 (Cramer's Rule)**.** *If A is invertible, the unique solution of $A\vec{x} = \vec{b}$ is given by*

$$x_i = \frac{\det A_i}{\det A}, \qquad i = 1, \ldots, n,$$

where A_i is A with its i-th column replaced by $\vec{b}$.

Proof. $\vec{x} = A^{-1}\vec{b} = \frac{1}{\det A}\operatorname{adj} A\,\vec{b}$, so $x_i = \frac{1}{\det A}\sum_k C_{ki}b_k$, and $\sum_k b_k C_{ki}$ is the expansion of $\det A_i$ along column i. □

Example 8.13. Solve $3x_1 + 2x_2 = 7$, $x_1 + 4x_2 = 9$. Here $\det A = 12 - 2 = 10$,

$$x_1 = \frac{1}{10}\begin{vmatrix}7 & 2\\ 9 & 4\end{vmatrix} = \frac{28-18}{10} = 1, \qquad x_2 = \frac{1}{10}\begin{vmatrix}3 & 7\\ 1 & 9\end{vmatrix} = \frac{27-7}{10} = 2.$$

Check: $3 + 4 = 7$, $1 + 8 = 9$.

Cramer's rule is mostly of theoretical value (e.g. it shows the solution depends continuously on the entries); for computation, row reduction is far cheaper.

8.6 Area and volume

Theorem 8.14. *If $\vec{u}, \vec{v} \in \mathbb{R}^2$, the parallelogram they span has area $|\det[\vec{u}\ \vec{v}]|$. If $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$, the parallelepiped they span has volume $|\det[\vec{u}\ \vec{v}\ \vec{w}]| = |\vec{u} \cdot (\vec{v} \times \vec{w})|$. More generally, a linear map $L : \mathbb{R}^n \to \mathbb{R}^n$ multiplies n-dimensional volumes by $|\det[L]|$.*

Idea. Area/volume is unchanged by shears (adding a multiple of one column to another), scales by $|c|$ when one edge is scaled by c, and is 1 for the unit square/cube; the determinant behaves the same way (up to sign). □

Example 8.15. The parallelogram spanned by $(3, 1)$ and $(1, 2)$ has area $\left|\det\begin{bmatrix}3 & 1\\ 1 & 2\end{bmatrix}\right| = |6 - 1| = 5$.

The parallelepiped spanned by $(1, 2, 0)$, $(0, 1, 3)$, $(1, 0, 1)$ has volume

$$\left|\det\begin{bmatrix}1 & 0 & 1\\ 2 & 1 & 0\\ 0 & 3 & 1\end{bmatrix}\right| = |1(1 \cdot 1 - 0 \cdot 3) - 0 + 1(2 \cdot 3 - 1 \cdot 0)| = |1 + 6| = 7.$$

The sign of the determinant records orientation: a negative determinant means the map reverses orientation (like a reflection).

Summary

- $\det A$ by cofactor expansion along any row or column; triangular: product of diagonal.
- Row operations: scaling by c multiplies det by c; swap negates; adding a multiple of a row changes nothing.
- $\det(AB) = \det A \det B$, $\det A^T = \det A$, $\det A^{-1} = 1/\det A$, $\det(cA) = c^n \det A$.
- A invertible $\iff \det A \neq 0$; $A^{-1} = \frac{1}{\det A}\operatorname{adj} A$; Cramer's rule $x_i = \det A_i / \det A$.
- $|\det|$ = area/volume scaling factor.

Chapter 9

Eigenvalues and Eigenvectors

9.1 Definitions

Many linear operators act very simply in certain directions: a projection fixes its line and kills the orthogonal line; a reflection fixes its line and flips the orthogonal one. Such special directions are eigenvectors.

Definition 9.1. Let A be $n \times n$. A scalar λ is an *eigenvalue* of A if there is a *non-zero* vector $\vec{v}$ with $A\vec{v} = \lambda\vec{v}$; such a $\vec{v}$ is an *eigenvector* of A corresponding to λ. The same terms apply to a linear operator L via its standard matrix.

Pitfall

$\vec{0}$ is never an eigenvector (since $A\vec{0} = \lambda\vec{0}$ for every λ), but 0 *can* be an eigenvalue: 0 is an eigenvalue of A iff $\text{Null}(A) \neq \{\vec{0}\}$ iff A is not invertible.

Example 9.2. For $\text{proj}_{\vec{d}}$ on $\mathbb{R}^2$: $\vec{d}$ is an eigenvector with eigenvalue 1, and any non-zero vector orthogonal to $\vec{d}$ is an eigenvector with eigenvalue 0. For $\text{refl}_{\vec{d}}$ the eigenvalues are 1 and -1. A rotation R_θ with $0 < \theta < \pi$ has no real eigenvectors at all: no non-zero vector keeps its direction.

9.2 The characteristic polynomial

$A\vec{v} = \lambda\vec{v}$ with $\vec{v} \neq \vec{0}$ means $(A - \lambda I)\vec{v} = \vec{0}$ has a non-trivial solution, i.e. $A - \lambda I$ is not invertible, i.e. $\det(A - \lambda I) = 0$.

Definition 9.3. The *characteristic polynomial* of A is $C(\lambda) = \det(A - \lambda I)$, a polynomial of degree n in λ (with leading coefficient $(-1)^n$).

Theorem 9.4. *λ is an eigenvalue of A iff $C(\lambda) = 0$. The eigenvectors for λ are the non-zero vectors of* $\text{Null}(A - \lambda I)$.

Definition 9.5 (Eigenspace, multiplicities)**.** For an eigenvalue λ, the *eigenspace* is $E_\lambda = \text{Null}(A - \lambda I)$, a subspace of $\mathbb{R}^n$. The *algebraic multiplicity* a_λ is the multiplicity of λ as a root of C; the *geometric multiplicity* is $g_\lambda = \dim E_\lambda$.

Algorithm

(1) Compute $C(\lambda) = \det(A - \lambda I)$ and factor it. (2) For each root λ, row reduce $A - \lambda I$ and find a basis of its null space. (Since λ is a root, $A - \lambda I$ *must* have a non-trivial null space; if your row reduction gives only $\vec{0}$, you made an arithmetic error.)

Example 9.6. Let $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. $C(\lambda) = \begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5)$.

- $\lambda = 2$: $A - 2I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{2}R_1} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{bmatrix}$, so $x_1 = -\frac{1}{2}t$; taking $t = -2$, $E_2 = \text{Span}\{(1, -2)\}$.
- $\lambda = 5$: $A - 5I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \xrightarrow{R_2 + 2R_1} \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \xrightarrow{-R_1} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$, so $E_5 = \text{Span}\{(1, 1)\}$.

Check: $A(1, -2) = (4 - 2, 2 - 6) = (2, -4) = 2(1, -2)$ and $A(1, 1) = (5, 5) = 5(1, 1)$. Here $a_\lambda = g_\lambda = 1$ for both.

Example 9.7 (Repeated eigenvalue)**.** Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Expanding along the first row,

$$C(\lambda) = \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = -\lambda(\lambda^2 - 1) - 1(-\lambda - 1) + 1(1 + \lambda) = -\lambda^3 + 3\lambda + 2 = -(\lambda + 1)^2(\lambda - 2).$$

(Check: $(\lambda + 1)^2(\lambda - 2) = (\lambda^2 + 2\lambda + 1)(\lambda - 2) = \lambda^3 - 3\lambda - 2$.)

- $\lambda = -1$ ($a_{-1} = 2$): $A + I = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. With $x_2 = s$, $x_3 = t$: $E_{-1} = \text{Span}\{(-1, 1, 0), (-1, 0, 1)\}$, so $g_{-1} = 2$.
- $\lambda = 2$ ($a_2 = 1$):

$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -2 & 1 \\ -2 & 1 & 1 \\ 1 & 1 & -2 \end{bmatrix} \xrightarrow[R_3 - R_1]{R_2 + 2R_1} \begin{bmatrix} 1 & -2 & 1 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{bmatrix}$$

$$\xrightarrow[-\frac{1}{3}R_2]{R_3 + R_2} \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 + 2R_2} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix},$$

so $E_2 = \text{Span}\{(1, 1, 1)\}$, $g_2 = 1$. Check: $A(1, 1, 1) = (2, 2, 2)$.

Example 9.8 (Geometric multiplicity smaller than algebraic)**.** $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ has $C(\lambda) = (2 - \lambda)^2$, so $\lambda = 2$ with $a_2 = 2$. But $A - 2I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has rank 1, so $E_2 = \text{Span}\{(1, 0)\}$ and $g_2 = 1 < 2$.

Example 9.9 (No real eigenvalues). $[R_{\pi/2}] = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ has $C(\lambda) = \lambda^2 + 1$, with no real roots. Over $\mathbb{C}$ it has eigenvalues $\pm i$. (In Math 136 we mostly work over $\mathbb{R}$; complex eigenvalues are treated in Math 235.)

9.3 Properties

Theorem 9.10. *If A is upper or lower triangular, its eigenvalues are its diagonal entries (with multiplicity).*

Proof. $A - \lambda I$ is triangular, so $C(\lambda) = (a_{11} - \lambda) \cdots (a_{nn} - \lambda)$. □

Theorem 9.11. *For every eigenvalue λ, $1 \le g_\lambda \le a_\lambda$.*

Proof sketch. $g_\lambda \ge 1$ since an eigenvalue has an eigenvector. For the upper bound, extend a basis $\{\vec{v}_1, \ldots, \vec{v}_g\}$ of E_λ to a basis of $\mathbb{R}^n$ and let P have these as columns. Then $P^{-1}AP = \begin{bmatrix} \lambda I_g & B \\ O & D \end{bmatrix}$, which has characteristic polynomial $(\lambda - t)^g \det(D - tI)$ (in variable t). Similar matrices share characteristic polynomials (Chapter 10), so $(\lambda - t)^g$ divides $C(t)$. □

Theorem 9.12. *If $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A listed with algebraic multiplicity (over $\mathbb{C}$), then*

$$\lambda_1 + \cdots + \lambda_n = \operatorname{tr} A \quad (\textit{sum of diagonal entries}), \qquad \lambda_1 \cdots \lambda_n = \det A.$$

Proof sketch. $C(\lambda) = (-1)^n(\lambda - \lambda_1) \cdots (\lambda - \lambda_n)$. Setting $\lambda = 0$ gives $\det A = C(0) = \lambda_1 \cdots \lambda_n$. Comparing coefficients of λ^{n-1} (only the product of the diagonal entries contributes) gives the trace statement. □

Check on the examples: $2 + 5 = 7 = 4 + 3$ and $2 \cdot 5 = 10 = 12 - 2$; and $-1 - 1 + 2 = 0 = \operatorname{tr} A$, $(-1)(-1)(2) = 2 = \det A$ for the 3×3 example.

Theorem 9.13. *If $\vec{v}$ is an eigenvector of A for λ, then $\vec{v}$ is an eigenvector of A^k for λ^k, and (if A is invertible) of A^{-1} for λ^{-1}.*

Proof. $A^2\vec{v} = A(\lambda\vec{v}) = \lambda^2\vec{v}$, and induct. If A is invertible then $\lambda \neq 0$, and applying A^{-1} to $A\vec{v} = \lambda\vec{v}$ gives $\vec{v} = \lambda A^{-1}\vec{v}$. □

Theorem 9.14. *Eigenvectors corresponding to distinct eigenvalues are linearly independent. More generally, if $\lambda_1, \ldots, \lambda_k$ are distinct and B_i is a basis of E_{λ_i}, then $B_1 \cup \cdots \cup B_k$ is linearly independent.*

Proof (first statement). Suppose not, and let $\{\vec{v}_1, \ldots, \vec{v}_m\}$ be a shortest dependent list of such eigenvectors, so $c_1\vec{v}_1 + \cdots + c_m\vec{v}_m = \vec{0}$ with all $c_i \neq 0$ (by minimality). Apply $A - \lambda_m I$: $c_1(\lambda_1 - \lambda_m)\vec{v}_1 + \cdots + c_{m-1}(\lambda_{m-1} - \lambda_m)\vec{v}_{m-1} = \vec{0}$, a shorter non-trivial relation (the coefficients are non-zero because the λ_i are distinct). Contradiction. □

Summary

- $A\vec{v} = \lambda\vec{v}$, $\vec{v} \neq \vec{0}$. Eigenvalues are roots of $C(\lambda) = \det(A - \lambda I)$; eigenspaces are $E_\lambda = \operatorname{Null}(A - \lambda I)$.
- $1 \le g_\lambda \le a_\lambda$; the sum of eigenvalues is the trace and the product is the determinant.

- Triangular matrices: eigenvalues on the diagonal. 0 is an eigenvalue iff A is singular.
- Eigenvectors from distinct eigenvalues are linearly independent.

Chapter 10

Diagonalization

10.1 Similar matrices

Definition 10.1. $n \times n$ matrices A and B are *similar* if there is an invertible P with $P^{-1}AP = B$.

Similarity is an equivalence relation (reflexive with $P = I$; symmetric with P^{-1}; transitive since $(PQ)^{-1}A(PQ) = Q^{-1}(P^{-1}AP)Q$). In Chapter 12 we will see that similar matrices represent the same linear operator in different bases, which explains why they share so much.

Theorem 10.2. *If A and B are similar, then they have the same characteristic polynomial, hence the same eigenvalues with the same algebraic multiplicities, and the same determinant, trace and rank.*

Proof. $\det(B-\lambda I) = \det(P^{-1}AP-\lambda P^{-1}P) = \det\left(P^{-1}(A-\lambda I)P\right) = \det(P^{-1})\det(A-\lambda I)\det P = \det(A-\lambda I)$. Determinant and trace are (up to sign) coefficients of the characteristic polynomial. For rank, multiplying by invertible matrices does not change the null space dimension: $B\vec{x} = \vec{0} \iff A(P\vec{x}) = \vec{0}$, so $\operatorname{Null} B = P^{-1}\operatorname{Null} A$. □

Pitfall

The converse fails: I and $\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ have the same characteristic polynomial $(1-\lambda)^2$ but are not similar (the only matrix similar to I is $P^{-1}IP = I$).

10.2 The Diagonalization Theorem

Definition 10.3. A is *diagonalizable* if it is similar to a diagonal matrix D: $P^{-1}AP = D$. We say P *diagonalizes* A.

Theorem 10.4 (Diagonalization Theorem)**.** *An $n \times n$ matrix A is diagonalizable (over $\mathbb{R}$) if and only if there is a basis of $\mathbb{R}^n$ consisting of eigenvectors of A. In that case, if $\{\vec{v}_1, \ldots, \vec{v}_n\}$ is such a basis with $A\vec{v}_i = \lambda_i\vec{v}_i$, then*

$$P = [\vec{v}_1 \ \cdots \ \vec{v}_n], \qquad P^{-1}AP = D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Proof. $P^{-1}AP = D$ iff $AP = PD$ (with P invertible). Column-by-column, $AP = [A\vec{v}_1 \cdots A\vec{v}_n]$ and $PD = [\lambda_1\vec{v}_1 \cdots \lambda_n\vec{v}_n]$. So $AP = PD$ iff each $\vec{v}_i$ satisfies $A\vec{v}_i = \lambda_i\vec{v}_i$; and P is invertible iff its columns form a basis of $\mathbb{R}^n$ (Invertible Matrix Theorem). Since basis vectors are non-zero, they are eigenvectors. □

Theorem 10.5 (Multiplicity test)**.** *A is diagonalizable over $\mathbb{R}$ if and only if all roots of $C(\lambda)$ are real and $g_\lambda = a_\lambda$ for every eigenvalue λ. In particular, an $n \times n$ matrix with n distinct real eigenvalues is diagonalizable.*

Proof. Combining bases of the eigenspaces gives a linearly independent set (Chapter 9) with $\sum g_\lambda$ vectors. Since $g_\lambda \le a_\lambda$ and $\sum a_\lambda \le n$ (with equality iff all roots are real), we get n vectors exactly when all roots are real and $g_\lambda = a_\lambda$ for all λ. Conversely, any eigenbasis contains at most g_λ vectors from each E_λ. □

Algorithm

1. Find and factor $C(\lambda)$. If some root is non-real, A is not diagonalizable over $\mathbb{R}$.
2. For each eigenvalue find a basis of E_λ. If some $g_\lambda < a_\lambda$, stop: not diagonalizable.
3. Put the eigenvectors as the columns of P and the corresponding eigenvalues, *in the same order*, on the diagonal of D.

Example 10.6. For $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ (Chapter 9) we found $E_2 = \text{Span}\{(1,-2)\}$ and $E_5 = \text{Span}\{(1,1)\}$. So

$$P = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}, \quad P^{-1} = \frac{1}{3}\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}.$$

Check $AP = \begin{bmatrix} 4-2 & 4+1 \\ 2-6 & 2+3 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ -4 & 5 \end{bmatrix}$ and $PD = \begin{bmatrix} 2 & 5 \\ -4 & 5 \end{bmatrix}$. Equal, so $P^{-1}AP = D$.

Example 10.7. For $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ we found $a_{-1} = g_{-1} = 2$ and $a_2 = g_2 = 1$, so A is diagonalizable with

$$P = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

(Check one column: $A(-1,0,1) = (1,0,-1) = -1 \cdot (-1,0,1)$.)

Example 10.8 (Not diagonalizable)**.** $A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ is upper triangular, so its eigenvalues are 3 ($a_3 = 2$) and 5 ($a_5 = 1$). $A - 3I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ has rank 2, so $g_3 = 3 - 2 = 1 < 2$. Not diagonalizable.

10.3 Powers of matrices

If $A = PDP^{-1}$, then $A^k = (PDP^{-1})(PDP^{-1})\cdots(PDP^{-1}) = PD^kP^{-1}$ because the inner $P^{-1}P$ factors cancel; and D^k is computed entrywise on the diagonal.

Example 10.9. For $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$:

$$A^k = PD^kP^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2^k & 0 \\ 0 & 5^k \end{bmatrix} \frac{1}{3} \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2^k & 5^k \\ -2^{k+1} & 5^k \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 2^k + 2 \cdot 5^k & 5^k - 2^k \\ 2 \cdot 5^k - 2^{k+1} & 5^k + 2^{k+1} \end{bmatrix}.$$

Check $k = 1$: $\frac{1}{3} \begin{bmatrix} 2+10 & 5-2 \\ 10-4 & 5+4 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$. Check $k = 0$: $\frac{1}{3} \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = I$.

Example 10.10 (A Markov chain)**.** Each year 10% of city residents move to the suburbs and 20% of suburban residents move to the city. If $\vec{x}_k = (c_k, s_k)$ are the population fractions in year k, then $\vec{x}_{k+1} = T\vec{x}_k$ with $T = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$. Its characteristic polynomial is $(0.9-\lambda)(0.8-\lambda) - 0.02 = \lambda^2 - 1.7\lambda + 0.7 = (\lambda - 1)(\lambda - 0.7)$. Eigenvectors: $T - I = \begin{bmatrix} -0.1 & 0.2 \\ 0.1 & -0.2 \end{bmatrix}$ gives $(2, 1)$; $T - 0.7I = \begin{bmatrix} 0.2 & 0.2 \\ 0.1 & 0.1 \end{bmatrix}$ gives $(1, -1)$. Writing $\vec{x}_0 = a(2,1) + b(1,-1)$,

$$\vec{x}_k = T^k\vec{x}_0 = a(2,1) + b(0.7)^k(1,-1) \longrightarrow a(2,1).$$

Since $c_k + s_k = 1$ always, and the $(1, -1)$ term contributes 0 to the sum, $3a = 1$; the long-run state is $\left(\frac{2}{3}, \frac{1}{3}\right)$ regardless of the starting distribution.

Diagonalization is also used to solve linear recurrences: the Fibonacci numbers satisfy $\begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_k \\ F_{k-1} \end{bmatrix}$, and diagonalizing this matrix (eigenvalues $\frac{1 \pm \sqrt{5}}{2}$) gives Binet's closed formula.

Summary

- Similar matrices ($B = P^{-1}AP$) share characteristic polynomial, eigenvalues, determinant, trace, rank.
- A is diagonalizable iff $\mathbb{R}^n$ has a basis of eigenvectors iff all eigenvalues are real with $g_\lambda = a_\lambda$. Distinct eigenvalues suffice.
- $P = [\text{eigenvectors}]$, $D = \text{diag}(\text{eigenvalues in the same order})$, $AP = PD$.
- $A^k = PD^kP^{-1}$.

Chapter 11

Vector Spaces

11.1 The axioms

Polynomials, matrices and functions can be added and scaled just like vectors in $\mathbb{R}^n$, and all the theory of span, independence and bases carries over. We abstract the ten properties of Chapter 1.

Definition 11.1 (Vector space)**.** A *(real) vector space* is a set $\mathbb{V}$ with an operation of addition $\vec{x}+\vec{y}$ and scalar multiplication $c\vec{x}$ ($c \in \mathbb{R}$) such that for all $\vec{x}, \vec{y}, \vec{z} \in \mathbb{V}$ and $c, d \in \mathbb{R}$:

V1. $\vec{x}+\vec{y} \in \mathbb{V}$;

V2. $(\vec{x}+\vec{y})+\vec{z} = \vec{x}+(\vec{y}+\vec{z})$;

V3. $\vec{x}+\vec{y} = \vec{y}+\vec{x}$;

V4. there is an element $\vec{0} \in \mathbb{V}$ (the *zero vector*) with $\vec{x}+\vec{0} = \vec{x}$ for all $\vec{x}$;

V5. for each $\vec{x}$ there is $-\vec{x} \in \mathbb{V}$ (the *additive inverse*) with $\vec{x}+(-\vec{x}) = \vec{0}$;

V6. $c\vec{x} \in \mathbb{V}$;

V7. $c(d\vec{x}) = (cd)\vec{x}$;

V8. $(c+d)\vec{x} = c\vec{x}+d\vec{x}$;

V9. $c(\vec{x}+\vec{y}) = c\vec{x}+c\vec{y}$;

V10. $1\vec{x} = \vec{x}$.

Elements of $\mathbb{V}$ are called *vectors*, whatever they actually are.

Example 11.2 (Standard examples)**.**

- $\mathbb{R}^n$.
- $P_n(\mathbb{R})$: polynomials $a_0 + a_1x + \cdots + a_nx^n$ of degree at most n, with the usual addition and scalar multiplication of polynomials. The zero vector is the zero polynomial.
- $M_{m\times n}(\mathbb{R})$: $m \times n$ matrices with entrywise operations; zero vector $O_{m,n}$.
- $\mathcal{F}(a,b)$: all functions $f : (a,b) \to \mathbb{R}$, with $(f+g)(x) = f(x)+g(x)$ and $(cf)(x) = cf(x)$. The zero vector is the function that is identically 0. Subspaces include continuous functions $C(a,b)$ and differentiable functions.

- The trivial vector space $\{\vec{0}\}$.

Example 11.3 (A strange vector space)**.** Let $\mathbb{V} = \{x \in \mathbb{R} : x > 0\}$ with "addition" $x \oplus y = xy$ and "scalar multiplication" $c \odot x = x^c$. Then $\mathbb{V}$ is a vector space: e.g. V4 holds with zero vector 1 (since $x \oplus 1 = x$), V5 with $-x = 1/x$, and V8 is $x^{c+d} = x^c x^d = (c \odot x) \oplus (d \odot x)$. The axioms only care about how the operations behave, not what they look like.

Theorem 11.4. *In any vector space:* $0\vec{x} = \vec{0}$, $(-1)\vec{x} = -\vec{x}$, $c\vec{0} = \vec{0}$*; the zero vector and additive inverses are unique.*

Proof of $0\vec{x} = \vec{0}$. $0\vec{x} = (0+0)\vec{x} = 0\vec{x} + 0\vec{x}$ by V8. Adding $-(0\vec{x})$ to both sides and using V2, V5, V4 gives $\vec{0} = 0\vec{x}$. □

11.2 Subspaces

Definition 11.5. A subset $\mathbb{S}$ of a vector space $\mathbb{V}$ is a *subspace* if it is a vector space under the same operations.

Theorem 11.6 (Subspace Test)**.** $\mathbb{S} \subseteq \mathbb{V}$ *is a subspace iff* $\mathbb{S}$ *is non-empty and closed under addition and scalar multiplication.*

Example 11.7. The symmetric matrices $\mathbb{S} = \{A \in M_{2\times 2}(\mathbb{R}) : A^T = A\}$ form a subspace: $O^T = O$; and $(A+B)^T = A^T + B^T = A + B$, $(cA)^T = cA^T = cA$.

Example 11.8. $\mathbb{S} = \{p \in P_2(\mathbb{R}) : p(1) = 0\}$ is a subspace: the zero polynomial vanishes at 1, and $(p+q)(1) = 0 + 0$, $(cp)(1) = c \cdot 0$. But $\{p \in P_2(\mathbb{R}) : p(1) = 1\}$ is not (it does not contain 0), and neither is the set of polynomials of degree *exactly* 2 ($x^2 + (-x^2) = 0$ has degree less than 2).

Example 11.9. The invertible 2×2 matrices do not form a subspace of $M_{2\times 2}(\mathbb{R})$: I and $-I$ are invertible but $I + (-I) = O$ is not.

11.3 Span and linear independence

Definitions of linear combination, Span, linear (in)dependence, spanning set and basis are word-for-word the same as in $\mathbb{R}^n$. Span is always a subspace. The difference is computational: we must translate problems about polynomials or matrices into systems of equations by comparing coefficients.

Example 11.10. Is $\{1+x+x^2,\ 1+2x+3x^2,\ 2+3x+4x^2\}$ linearly independent in $P_2(\mathbb{R})$? Suppose $c_1(1+x+x^2) + c_2(1+2x+3x^2) + c_3(2+3x+4x^2) = 0$. Comparing the coefficients of $1, x, x^2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 1 & 2 & 3 & 0 \\ 1 & 3 & 4 & 0 \end{array}\right] \xrightarrow[R_3-R_1]{R_2-R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 2 & 0 \end{array}\right] \xrightarrow{R_3-2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

c_3 is free, so there are non-trivial solutions: the set is dependent. Taking $c_3 = -1$: $c_2 = 1$, $c_1 = 1$, i.e. $(1+x+x^2) + (1+2x+3x^2) = 2+3x+4x^2$.

Example 11.11. Is $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ in Span $\left\{\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}\right\}$? We need $c_1 + c_3 = 1$, $c_2 + c_3 = 2$, $c_1 = 3$, $c_2 = 4$ (entries $(1,1), (1,2), (2,1), (2,2)$). Then $c_3 = 1 - 3 = -2$ and $c_3 = 2 - 4 = -2$: consistent. So yes, with $(c_1, c_2, c_3) = (3, 4, -2)$.

Example 11.12 (Functions)**.** $\{\sin x, \cos x\}$ is linearly independent in $\mathcal{F}(\mathbb{R})$: if $c_1 \sin x + c_2 \cos x = 0$ for *all* x, then $x = 0$ gives $c_2 = 0$ and $x = \pi/2$ gives $c_1 = 0$. (Note that the equation must hold as functions, i.e. at every x.)

11.4 Bases and dimension

Theorem 11.13 (Replacement / Exchange Lemma)**.** *If* $\{\vec{v}_1, \ldots, \vec{v}_n\}$ *spans* $\mathbb{V}$ *and* $\{\vec{u}_1, \ldots, \vec{u}_k\} \subseteq \mathbb{V}$ *is linearly independent, then* $k \leq n$.

Proof sketch. Write $\vec{u}_j = \sum_i a_{ij}\vec{v}_i$. If $k > n$, the homogeneous system $A\vec{c} = \vec{0}$ ($A = [a_{ij}]$ is $n \times k$) has a non-trivial solution, and then $\sum_j c_j\vec{u}_j = \sum_i(\sum_j a_{ij}c_j)\vec{v}_i = \vec{0}$, contradicting independence. □

Definition 11.14 (Dimension)**.** If $\mathbb{V}$ has a basis with n vectors, then (by the lemma) every basis has n vectors, and $\dim \mathbb{V} = n$; $\mathbb{V}$ is *finite-dimensional*. $\dim\{\vec{0}\} = 0$. If no finite set spans $\mathbb{V}$, it is *infinite-dimensional* (e.g. $\mathcal{F}(\mathbb{R})$, or the space of all polynomials).

Example 11.15. $\dim \mathbb{R}^n = n$; $\dim P_n(\mathbb{R}) = n+1$ (standard basis $\{1, x, \ldots, x^n\}$); $\dim M_{m\times n}(\mathbb{R}) = mn$ (standard basis $\{E_{ij}\}$ with a single 1 in position (i, j)). The symmetric 2×2 matrices have basis $\left\{\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}, \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}, \begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}\right\}$, dimension 3.

Example 11.16. Find a basis and the dimension of $\mathbb{S} = \{p \in P_2(\mathbb{R}) : p(1) = 0\}$. Writing $p = a+bx+cx^2$, the condition is $a+b+c = 0$, i.e. $a = -b-c$, so $p = -b-c+bx+cx^2 = b(x-1)+c(x^2-1)$. Thus $\mathbb{S} = \text{Span}\{x-1, x^2-1\}$, and this set is independent (neither is a multiple of the other). So $\dim \mathbb{S} = 2$.

Theorem 11.17. *Let* $\dim \mathbb{V} = n$.

1. *Any set of more than n vectors is dependent; any set of fewer than n vectors does not span.*
2. *A set of exactly n vectors is a basis iff it is independent iff it spans ("two out of three").*
3. *Every independent set can be extended to a basis; every spanning set contains a basis.*
4. *If* $\mathbb{S}$ *is a subspace,* $\dim \mathbb{S} \leq n$*, with equality iff* $\mathbb{S} = \mathbb{V}$.

Proof sketch of (3). If an independent set B does not span, pick $\vec{v} \notin \text{Span}\, B$; then $B \cup \{\vec{v}\}$ is still independent (a relation with non-zero coefficient on $\vec{v}$ would put $\vec{v}$ in $\text{Span}\, B$). This stops after at most n steps by (1). For a spanning set, repeatedly remove a vector that is a combination of the others (Chapter 1). □

Example 11.18. Extend $\{1 + x,\ 1 - x\}$ to a basis of $P_2(\mathbb{R})$. Since $\dim P_2(\mathbb{R}) = 3$, we need one more vector not in their span. Every combination $a(1 + x) + b(1 - x)$ has no x^2 term, so x^2 works: $\{1 + x, 1 - x, x^2\}$ is independent, hence a basis by "two out of three".

11.5 Coordinates

Definition 11.19. Let $\mathcal{B} = \{\vec{v}_1, \ldots, \vec{v}_n\}$ be an (ordered) basis of $\mathbb{V}$. For $\vec{x} \in \mathbb{V}$, write uniquely $\vec{x} = c_1\vec{v}_1 + \cdots + c_n\vec{v}_n$. The *coordinate vector* of $\vec{x}$ with respect to $\mathcal{B}$ is

$$[\vec{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n.$$

Theorem 11.20. *The coordinate map is linear:* $[b\vec{x}+c\vec{y}]_{\mathcal{B}} = b[\vec{x}]_{\mathcal{B}} + c[\vec{y}]_{\mathcal{B}}$. *Moreover* $\{\vec{x}_1,\dots,\vec{x}_k\}$ *is independent (spans* $\mathbb{V}$*) iff* $\{[\vec{x}_1]_{\mathcal{B}},\dots,[\vec{x}_k]_{\mathcal{B}}\}$ *is independent (spans* $\mathbb{R}^n$*).*

This is why comparing coefficients worked in the examples above: we were really computing in $\mathbb{R}^n$ via coordinates with respect to the standard basis.

Example 11.21. Find $[p]_{\mathcal{B}}$ for $p = 3+x+2x^2$ and $\mathcal{B} = \{1+x,\ 1-x,\ x^2\}$. Solve $a(1+x)+b(1-x)+cx^2 = 3+x+2x^2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow{R_2-R_1} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow{-\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \xrightarrow{R_1-R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right].$$

So $[p]_{\mathcal{B}} = (2,1,2)$. Check: $2(1+x)+(1-x)+2x^2 = 3+x+2x^2$.

Pitfall

A coordinate vector depends on the *order* of the basis vectors, and $[\vec{x}]_{\mathcal{B}}$ lives in $\mathbb{R}^n$ even when $\vec{x}$ is a polynomial or a matrix.

Summary

- A vector space is any set with + and scalar multiplication satisfying V1–V10: $\mathbb{R}^n$, $P_n(\mathbb{R})$, $M_{m\times n}(\mathbb{R})$, function spaces.
- Subspace test: non-empty, closed under + and scalar multiplication.
- Span, independence and bases work as in $\mathbb{R}^n$; reduce to linear systems by comparing coefficients.
- All bases have the same size $\dim\mathbb{V}$; $\dim P_n(\mathbb{R}) = n+1$, $\dim M_{m\times n}(\mathbb{R}) = mn$. With n vectors, independent $\iff$ spanning.
- Coordinates $[\vec{x}]_{\mathcal{B}} \in \mathbb{R}^n$ turn any n-dimensional space into $\mathbb{R}^n$.

Chapter 12

General Linear Maps and Change of Basis

12.1 Linear maps between vector spaces

Definition 12.1. Let $\mathbb{V}, \mathbb{W}$ be vector spaces. $L : \mathbb{V} \to \mathbb{W}$ is *linear* if $L(b\vec{x} + c\vec{y}) = bL(\vec{x}) + cL(\vec{y})$ for all $\vec{x}, \vec{y} \in \mathbb{V}$, $b, c \in \mathbb{R}$. Its *range* is $\text{Range}(L) = \{L(\vec{x}) : \vec{x} \in \mathbb{V}\}$ and its *kernel* is $\text{Ker}(L) = \{\vec{x} \in \mathbb{V} : L(\vec{x}) = \vec{0}\}$.

Example 12.2.

- Differentiation $D : P_n(\mathbb{R}) \to P_{n-1}(\mathbb{R})$, $D(p) = p'$.
- Transpose $T : M_{m\times n}(\mathbb{R}) \to M_{n\times m}(\mathbb{R})$, $T(A) = A^T$.
- Trace $\text{tr} : M_{n\times n}(\mathbb{R}) \to \mathbb{R}$.
- Evaluation $L : P_2(\mathbb{R}) \to \mathbb{R}^2$, $L(p) = (p(0), p(1))$.
- The coordinate map $\vec{x} \mapsto [\vec{x}]_{\mathcal{B}}$.

The determinant $\det : M_{n\times n}(\mathbb{R}) \to \mathbb{R}$ is *not* linear for $n \geq 2$ ($\det(2I) = 2^n \neq 2 \det I$).

As before, $\text{Ker}\, L$ is a subspace of $\mathbb{V}$, $\text{Range}\, L$ is a subspace of $\mathbb{W}$, and L is one-to-one iff $\text{Ker}\, L = \{\vec{0}\}$. The *rank* of L is $\dim \text{Range}\, L$ and the *nullity* is $\dim \text{Ker}\, L$.

Theorem 12.3 (Rank–Nullity)**.** *If $\mathbb{V}$ is finite-dimensional and $L : \mathbb{V} \to \mathbb{W}$ is linear, then* $\text{rank}\, L + \text{nullity}\, L = \dim \mathbb{V}$.

Proof. Let $\{\vec{v}_1, \ldots, \vec{v}_k\}$ be a basis of $\text{Ker}\, L$ and extend it to a basis $\{\vec{v}_1, \ldots, \vec{v}_k, \vec{v}_{k+1}, \ldots, \vec{v}_n\}$ of $\mathbb{V}$. We claim $\{L(\vec{v}_{k+1}), \ldots, L(\vec{v}_n)\}$ is a basis of $\text{Range}\, L$. Spanning: any $L(\vec{x}) = \sum c_i L(\vec{v}_i)$, and the terms with $i \leq k$ vanish. Independence: if $\sum_{i>k} c_i L(\vec{v}_i) = \vec{0}$ then $\sum_{i>k} c_i \vec{v}_i \in \text{Ker}\, L$, so it equals some $\sum_{i\leq k} d_i \vec{v}_i$; independence of the full basis forces all $c_i = 0$. Hence $\text{rank}\, L = n - k$. □

Example 12.4. For $D : P_3(\mathbb{R}) \to P_2(\mathbb{R})$: $\text{Ker}\, D$ = constants (dimension 1) and $\text{Range}\, D = P_2(\mathbb{R})$ (dimension 3); $1 + 3 = 4 = \dim P_3(\mathbb{R})$. For $\text{tr} : M_{2\times 2}(\mathbb{R}) \to \mathbb{R}$ (onto, rank 1), the kernel (trace-zero matrices) has dimension $4 - 1 = 3$.

12.2 The matrix of a linear map

Fix bases $\mathcal{B} = \{\vec{v}_1, \dots, \vec{v}_n\}$ of $\mathbb{V}$ and $\mathcal{C}$ of $\mathbb{W}$ (with $\dim \mathbb{W} = m$). Taking coordinates converts L into a map $\mathbb{R}^n \to \mathbb{R}^m$, which must be given by a matrix.

Definition 12.5. The *matrix of L with respect to $\mathcal{B}$ and $\mathcal{C}$* is the $m \times n$ matrix

$$ {}_{\mathcal{C}}[L]_{\mathcal{B}} = \Big[\, [L(\vec{v}_1)]_{\mathcal{C}} \;\; [L(\vec{v}_2)]_{\mathcal{C}} \;\; \cdots \;\; [L(\vec{v}_n)]_{\mathcal{C}} \,\Big]. $$

When $\mathbb{V} = \mathbb{W}$ and $\mathcal{C} = \mathcal{B}$ we write $[L]_{\mathcal{B}}$ (the *$\mathcal{B}$-matrix* of L).

Theorem 12.6. *For all $\vec{x} \in \mathbb{V}$: $[L(\vec{x})]_{\mathcal{C}} = {}_{\mathcal{C}}[L]_{\mathcal{B}}\,[\vec{x}]_{\mathcal{B}}$.*

Proof. If $[\vec{x}]_{\mathcal{B}} = (c_1, \dots, c_n)$ then $L(\vec{x}) = \sum c_i L(\vec{v}_i)$, so by linearity of coordinates $[L(\vec{x})]_{\mathcal{C}} = \sum c_i [L(\vec{v}_i)]_{\mathcal{C}}$, which is the matrix–vector product. □

When $\mathbb{V} = \mathbb{R}^n$, $\mathbb{W} = \mathbb{R}^m$ and both bases are standard, this is the standard matrix $[L]$ of Chapter 5.

Example 12.7. $D : P_2(\mathbb{R}) \to P_1(\mathbb{R})$ with standard bases $\{1, x, x^2\}$ and $\{1, x\}$: $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, so

$$ [D] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}. $$

For $p = 5 - 3x + 4x^2$: $[D]\,(5, -3, 4) = (-3, 8)$, i.e. $p' = -3 + 8x$. Correct.

Example 12.8. Let $L : P_2(\mathbb{R}) \to \mathbb{R}^2$, $L(p) = (p(0), p(1))$, with $\mathcal{B} = \{1, x, x^2\}$ and $\mathcal{C} = \{(1,0), (1,1)\}$. Then $L(1) = (1,1)$, $L(x) = (0,1)$, $L(x^2) = (0,1)$. In $\mathcal{C}$-coordinates, $a(1,0) + b(1,1) = (a+b, b)$, so $[(1,1)]_{\mathcal{C}} = (0,1)$ and $[(0,1)]_{\mathcal{C}} = (-1,1)$. Hence

$$ {}_{\mathcal{C}}[L]_{\mathcal{B}} = \begin{bmatrix} 0 & -1 & -1 \\ 1 & 1 & 1 \end{bmatrix}. $$

Check with $p = 2 + 3x + x^2$: ${}_{\mathcal{C}}[L]_{\mathcal{B}}(2, 3, 1) = (-4, 6)$, and $-4(1,0) + 6(1,1) = (2,6) = (p(0), p(1))$.

Example 12.9 (A well-chosen basis)**.** Let $L = \mathrm{proj}_{\vec{d}}$ on $\mathbb{R}^2$ with $\vec{d} = (1,2)$, and $\mathcal{B} = \{(1,2), (-2,1)\}$. Since $L(1,2) = (1,2)$ and $L(-2,1) = \vec{0}$,

$$ [L]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, $$

far simpler than the standard matrix $\frac{1}{5}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. Choosing a basis adapted to the geometry is exactly what diagonalization does.

12.3 Change of coordinates

Definition 12.10. Let $\mathcal{B} = \{\vec{v}_1, \dots, \vec{v}_n\}$ and $\mathcal{C}$ be bases of $\mathbb{V}$. The *change of coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$* is

$$ {}_{\mathcal{C}}P_{\mathcal{B}} = \Big[\, [\vec{v}_1]_{\mathcal{C}} \;\; \cdots \;\; [\vec{v}_n]_{\mathcal{C}} \,\Big], $$

i.e. the matrix of the identity map with respect to $\mathcal{B}$ and $\mathcal{C}$.

Theorem 12.11. *$[\vec{x}]_{\mathcal{C}} = {}_{\mathcal{C}}P_{\mathcal{B}}[\vec{x}]_{\mathcal{B}}$ for all $\vec{x}$, and ${}_{\mathcal{C}}P_{\mathcal{B}}$ is invertible with $({}_{\mathcal{C}}P_{\mathcal{B}})^{-1} = {}_{\mathcal{B}}P_{\mathcal{C}}$.*

Proof. The first statement is the previous theorem applied to the identity map. Then ${}_{\mathcal{B}}P_{\mathcal{C}}\,{}_{\mathcal{C}}P_{\mathcal{B}}[\vec{x}]_{\mathcal{B}} = [\vec{x}]_{\mathcal{B}}$ for every coordinate vector, so the product is I. □

In $\mathbb{R}^n$

If $\mathcal{S}$ is the standard basis and $\mathcal{B} = \{\vec{v}_1, \ldots, \vec{v}_n\}$, then ${}_{\mathcal{S}}P_{\mathcal{B}} = [\vec{v}_1 \ \cdots \ \vec{v}_n]$ (just write the basis vectors as columns), and ${}_{\mathcal{B}}P_{\mathcal{S}} = [\vec{v}_1 \ \cdots \ \vec{v}_n]^{-1}$.

Example 12.12. In $P_2(\mathbb{R})$ let $\mathcal{S} = \{1, x, x^2\}$ and $\mathcal{B} = \{1,\ 1+x,\ 1+x+x^2\}$. Then

$$ {}_{\mathcal{S}}P_{\mathcal{B}} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}. $$

Invert it:

$$ \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 - R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]. $$

(The last step subtracts the current R_2 and R_3 from R_1: $(1,1,1 \mid 1,0,0) - (0,1,0 \mid 0,1,-1) - (0,0,1 \mid 0,0,1) = (1,0,0 \mid 1,-1,0)$.) So ${}_{\mathcal{B}}P_{\mathcal{S}} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$. For $p = 3 + 2x + x^2$, $[p]_{\mathcal{B}} = {}_{\mathcal{B}}P_{\mathcal{S}}(3,2,1) = (1,1,1)$. Check: $1 + (1+x) + (1+x+x^2) = 3 + 2x + x^2$.

12.4 Change of basis for linear operators

Theorem 12.13. *Let $L : \mathbb{V} \to \mathbb{V}$ be linear and $\mathcal{B}, \mathcal{C}$ bases of $\mathbb{V}$, with $P = {}_{\mathcal{C}}P_{\mathcal{B}}$. Then*

$$ [L]_{\mathcal{B}} = P^{-1}[L]_{\mathcal{C}}P. $$

In particular, for an operator on $\mathbb{R}^n$ with $\mathcal{B} = \{\vec{v}_1, \ldots, \vec{v}_n\}$ and $P = [\vec{v}_1 \cdots \vec{v}_n]$, $[L]_{\mathcal{B}} = P^{-1}[L]P$.

Proof. For every $\vec{x}$: $P^{-1}[L]_{\mathcal{C}}P[\vec{x}]_{\mathcal{B}} = P^{-1}[L]_{\mathcal{C}}[\vec{x}]_{\mathcal{C}} = P^{-1}[L(\vec{x})]_{\mathcal{C}} = [L(\vec{x})]_{\mathcal{B}}$. Read right to left: convert to $\mathcal{C}$-coordinates, apply L, convert back. □

So matrices of the same operator in different bases are *similar*, and conversely similar matrices represent the same operator. Diagonalizing A means finding a basis $\mathcal{B}$ in which the operator $\vec{x} \mapsto A\vec{x}$ has a diagonal matrix, namely an eigenvector basis.

Example 12.14. Verify the projection example: with $P = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$, $P^{-1} = \frac{1}{5}\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$,

$$ P[L]_{\mathcal{B}}P^{-1} = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\frac{1}{5}\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}\frac{1}{5}\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = [L]. $$

12.5 Isomorphisms

Definition 12.15. A linear map $L : \mathbb{V} \to \mathbb{W}$ is an *isomorphism* if it is one-to-one and onto. Then $\mathbb{V}$ and $\mathbb{W}$ are *isomorphic*, written $\mathbb{V} \cong \mathbb{W}$.

An isomorphism has a linear inverse $L^{-1} : \mathbb{W} \to \mathbb{V}$, and it carries bases to bases, independent sets to independent sets, and subspaces to subspaces of the same dimension. Isomorphic spaces are "the same space with different names for the vectors".

Theorem 12.16. *Finite-dimensional vector spaces $\mathbb{V}, \mathbb{W}$ are isomorphic if and only if* $\dim \mathbb{V} = \dim \mathbb{W}$. *In particular, every n-dimensional space is isomorphic to $\mathbb{R}^n$ via $\vec{x} \mapsto [\vec{x}]_{\mathcal{B}}$.*

Proof. If L is an isomorphism, $\operatorname{nullity} L = 0$ and $\operatorname{rank} L = \dim \mathbb{W}$, so rank–nullity gives $\dim \mathbb{V} = \dim \mathbb{W}$. Conversely, if both have dimension n with bases $\{\vec{v}_i\}$ and $\{\vec{w}_i\}$, define $L(\sum c_i \vec{v}_i) = \sum c_i \vec{w}_i$; it is linear, onto (hits every combination of the $\vec{w}_i$) and one-to-one (the kernel is trivial by independence of the $\vec{w}_i$). □

Theorem 12.17. *If $\dim \mathbb{V} = \dim \mathbb{W} = n$ and $L : \mathbb{V} \to \mathbb{W}$ is linear, then L is one-to-one iff L is onto iff L is an isomorphism iff ${}_{\mathcal{C}}[L]_{\mathcal{B}}$ is invertible (for any bases).*

Proof. By rank–nullity, $\operatorname{nullity} L = 0$ iff $\operatorname{rank} L = n = \dim \mathbb{W}$. □

Example 12.18. $P_3(\mathbb{R}) \cong M_{2\times 2}(\mathbb{R}) \cong \mathbb{R}^4$ (all dimension 4); e.g. $a + bx + cx^2 + dx^3 \mapsto \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. But $P_2(\mathbb{R}) \ncong \mathbb{R}^4$. The map $L : P_2(\mathbb{R}) \to \mathbb{R}^3$, $L(p) = (p(0), p(1), p(2))$ is an isomorphism: its kernel consists of polynomials of degree at most 2 with three roots, so only the zero polynomial; one-to-one between spaces of the same dimension implies onto. (This is the reason quadratic interpolation through three points always works.)

Summary

- Linear maps between general spaces have kernels and ranges; $\operatorname{rank} L + \operatorname{nullity} L = \dim \mathbb{V}$.
- ${}_{\mathcal{C}}[L]_{\mathcal{B}}$ has columns $[L(\vec{v}_i)]_{\mathcal{C}}$ and satisfies $[L(\vec{x})]_{\mathcal{C}} = {}_{\mathcal{C}}[L]_{\mathcal{B}}[\vec{x}]_{\mathcal{B}}$.
- ${}_{\mathcal{C}}P_{\mathcal{B}}$ has columns $[\vec{v}_i]_{\mathcal{C}}$; its inverse is ${}_{\mathcal{B}}P_{\mathcal{C}}$.
- $[L]_{\mathcal{B}} = P^{-1}[L]_{\mathcal{C}}P$: matrices of one operator in different bases are similar.
- Isomorphism = bijective linear map; finite-dimensional spaces are isomorphic iff same dimension.